Sindh Science Learning Portal

Science

Class 5








Following the Sindh Textbook Board curriculum

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The Professional Educator · EduPulse Nexus

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Note: these lessons become final after review by teachers.

CHAPTER 1

Living Things and Classification

What we will learn Learn how living things are different from non-living things and how we sort them into groups.

1 What is a living thing?

Look around you. You can see **living things** like people, birds, trees and ants. You can also see **non-living things** like stones, books and chairs.

All living things share some features:

- They **grow** bigger with time.
- They **breathe** (take in air).
- They need **food** and **water**.
- They **move** on their own. Plants move slowly too, for example a sunflower turns towards the sun.
- They **reproduce** (make young ones like themselves).
- They **respond** to things around them, like heat, light and touch.

2 Non-living things

Non-living things do not grow, breathe, eat or reproduce on their own.

Some non-living things are **natural**, like stones, water, air and sand. Others are **made by people**, like pencils, bags, cars and fans.

A car moves, but it is not living. It moves only when a person drives it and it has petrol. It cannot grow or make baby cars!

3 Groups of living things: plants and animals

Scientists sort living things into groups. This is called **classification**. It makes them easier to study.

The two biggest groups are **plants** and **animals**.

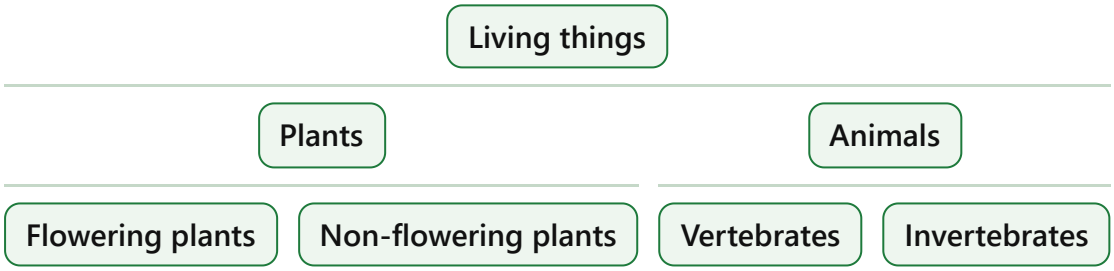
- **Plants** make their own food using sunlight. They cannot walk from place to place.
- **Animals** cannot make their own food. They eat plants or other animals. Most animals can move from place to place.

Plants can be **flowering plants** (like mango, rose and wheat) or **non-flowering plants** (like ferns and mosses).



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Diagram: the parts and what they do



Living things (جاندار · جاندار)
Things that grow, breathe, need food and water, and reproduce.

Plants (پودے · ہوتا)
Living things that make their own food using sunlight.

Animals (جانور · جانور)
Living things that eat plants or other animals. Most can move from place to place.

Flowering plants (گل وارا ہوتا · پھول دار پودے)
Plants that have flowers, fruits and seeds. Examples: mango, rose, wheat.

Non-flowering plants (بنا گلن وارا ہوتا · بغیر پھول والے پودے)
Plants that do not have flowers. Examples: fern, moss.

Vertebrates (ریڑھ دار جانور · فقاریہ)
Animals with a backbone. Examples: fish, frog, lizard, bird, cow.

Invertebrates (بنا ریڑھ وارا جانور · غیر فقاریہ)
Animals without a backbone. Examples: butterfly, snail, spider, ant.

4 Animals with and without a backbone

Animals are sorted by whether they have a **backbone** (a line of bones down the back).

- **Vertebrates** have a backbone. Examples: fish, frog, lizard, sparrow, cow and humans.
- **Invertebrates** do not have a backbone. Examples: butterfly, snail, spider, ant and mosquito.

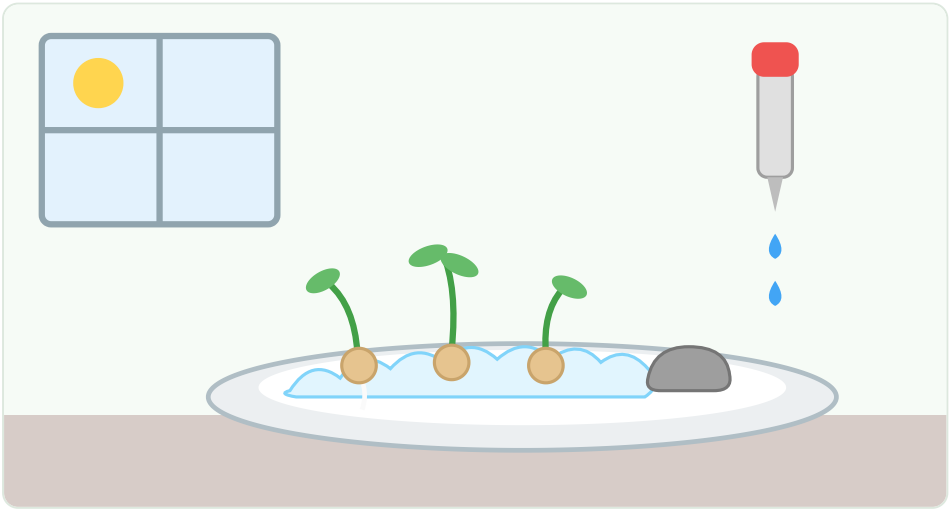
Vertebrates are put into five groups:

- **Fish** live in water and breathe with gills.
- **Amphibians** like frogs live both in water and on land.
- **Reptiles** like lizards and snakes have dry, scaly skin.
- **Birds** have feathers and lay eggs.
- **Mammals** like cows, cats and humans have hair and feed milk to their babies.

Key words

English	سنڌي	اردو
living thing	جاندار	جاندار
non-living thing	غير جاندار	بے جان
classification	درجه بندي	درجہ بندی
reproduce	ٻچا پيدا ڪرڻ	افزائشِ نسل
backbone	ريڑهه جو هڏو	ریڑھ کی ہڈی
vertebrates	ريڑهه دار جانور	فقاریہ
invertebrates	بنا ريڑهه وارا جانور	غیر فقاریہ
amphibians	ٻه جيوت جانور	جل تھلیے
reptiles	رينگڻ وارا جانور	رینگنے والے جانور
mammals	ڪير پيارڻ وارا جانور	ممالیہ

Experiment: Watch a seed come alive



You need

- 4 or 5 chickpea (chana) or moong seeds
- Some cotton wool
- A small plate
- Water
- A small stone

Steps

1. Put the cotton on the plate and make it wet with water.
2. Place the seeds and the small stone on the wet cotton.
3. Keep the plate near a window. Add a few drops of water every day so the cotton stays wet.
4. Look at the plate every day for one week. Draw what you see in your notebook.

Safety: Wash your hands after touching the seeds. Do not eat them.

What did we learn? After a few days the seeds grow small roots and leaves, but the stone does not change. Seeds are alive: they grow when they get water and air. A stone is non-living, so it never grows.

Practice questions

Circle the correct answer

1. **Which of these is a living thing?**

   (a) Stone (b) Chair
   (c) Tree (d) Book

2. **What do living things need to stay alive?**

   (a) Food and water (b) Paint
   (c) Plastic (d) Toys

3. **What do we call it when living things make young ones like themselves?**

   (a) Breathing (b) Reproduction
   (c) Moving (d) Sleeping

4. **Which of these is a non-living thing made by people?**

   (a) Pencil (b) Tree
   (c) Cat (d) Fish

5. **Which of these is a natural non-living thing?**

   (a) Bag (b) Sand
   (c) Fan (d) Car

6. **What is sorting living things into groups called?**

   (a) Classification (b) Breathing
   (c) Digestion (d) Melting

7. **Which living things make their own food?**

(a) Animals (b) Plants

(c) Birds (d) Insects

8. **Which animal has a backbone?**

(a) Butterfly (b) Snail

(c) Fish (d) Spider

9. **Which group of animals has feathers?**

(a) Fish (b) Birds

(c) Reptiles (d) Mammals

10. **A frog lives in water and on land. Which group is it in?**

(a) Amphibians (b) Reptiles

(c) Birds (d) Mammals

Write true or false

11. **A car is a living thing because it moves.** _____

12. **Plants are living things.** _____

13. **An ant does not have a backbone.** _____

14. **Mammals feed milk to their babies.** _____

Match the columns

15. Match each animal with its group.

1. Cow		(a) Bird
2. Sparrow		(b) Reptile
3. Snake		(c) Amphibian
4. Frog		(d) Mammal

Match the columns

16. Match each thing with its type.

1. Stone		(a) Man-made non-living thing
2. Pencil		(b) Plant
3. Mango tree		(c) Animal
4. Goat		(d) Natural non-living thing

Worksheet: Living Things and Classification

Name: _____ Date: _____

A. Fill in the blanks

Word box: classification · feathers · food · vertebrates · water

1. Living things need food and _____.
2. Sorting living things into groups is called _____.
3. Animals with a backbone are called _____.
4. Birds have _____ on their bodies.
5. Plants make their own _____ using sunlight.

B. Put ✓ for true and ✗ for false

1. ☐ A stone grows bigger every year.
2. ☐ Fish breathe with gills.
3. ☐ A snake is a mammal.
4. ☐ Animals cannot make their own food.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Butterfly	( ) Non-flowering plant
2. Rose	( ) Reptile
3. Fern	( ) Invertebrate
4. Lizard	( ) Flowering plant

D. Answer in short

1. Write any three features of living things.

2. A car moves. Why is it not a living thing?

3. Name the five groups of vertebrates.

E. Draw and label

Draw two living things and two non-living things. Write their names.

CHAPTER 2

Microorganisms

What we will learn Learn what microorganisms are, how bacteria, viruses and fungi differ, and how they can be useful or harmful.

1 What are microorganisms?

Some living things are so small that we cannot see them with our eyes. They are called **microorganisms** or **microbes**.

They can only be seen with a **microscope**. A microscope makes tiny things look much bigger.

Microorganisms are found everywhere: in soil, air and water, on plants and animals, and inside and outside our bodies.

There are three main groups: **bacteria**, **viruses** and **fungi**.

2 Bacteria, viruses and fungi

- **Bacteria**: living things made of one cell. Some are round, some are rod-shaped and some are spiral. They feed, breathe and grow by dividing.
- **Viruses**: the smallest microbes. They can only be seen with a very powerful microscope. A virus can only grow inside the cells of a living thing. Colds, flu, polio and measles are caused by viruses.
- **Fungi**: some fungi are tiny and have one cell, like **yeast**. Some are large, like **mushrooms**. The green or black **mould** on old bread is also a fungus.

Fungi cannot make their own food. They take food from dead or rotting things.



Picture: Jan Myšák · CC BY-SA 4.0 · Wikimedia Commons

Diagram: the parts and what they do



Microorganisms (خوردبيني جاندار · خوردبینی جاندار)
Tiny living things that can only be seen with a microscope.

Bacteria (بيكتيريا · بیکتیریا)
One-celled. Some make yoghurt; some cause cholera and typhoid.

Viruses (وائرس · وائرس)
The smallest. They grow only inside living cells. They cause colds, flu and polio.

Fungi (فنگس · فنجائی)
Yeast, mould and mushrooms. Yeast makes dough rise; penicillin comes from a fungus.

3 Useful microorganisms

Most microorganisms are not harmful. Many are our friends:

- **Decomposing**: bacteria and fungi break down dead plants, dead animals and rubbish and mix them into the **soil**. This makes the soil rich.
- **Making food**: **yeast** makes dough rise, so naan and bread become soft. Bacteria turn milk into **yoghurt** and cheese.
- **Digestion**: some bacteria living in our intestines help us digest food.
- **Medicines**: some **antibiotics** are made from microbes. **Penicillin** is made from a fungus. **Vaccines** (like polio drops) protect us from diseases.

4 Harmful microorganisms and staying safe

Some microbes **spoil** food and spread diseases. A disease caused by microbes is called an **infection**.

- Caused by bacteria: **cholera**, **typhoid** and some kinds of diarrhoea.
- Caused by viruses: colds, flu, measles and **dengue** (the dengue virus is spread by mosquitoes).
- Caused by fungi: athlete's foot (itching between the toes) and some skin diseases.

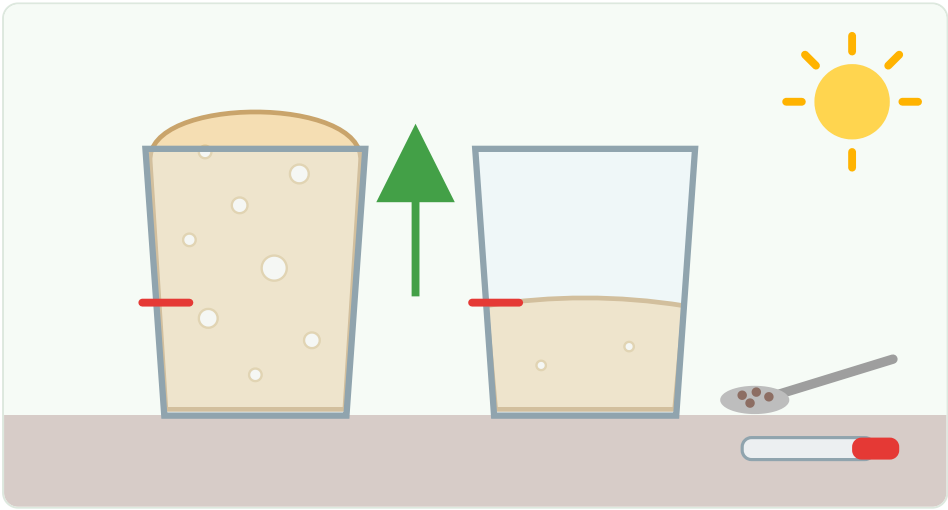
Microbes enter the body through the mouth (dirty food and water), the nose (breathing) and cuts in the skin.

Stay safe: **wash your hands** with soap, boil drinking water, keep food covered, keep your surroundings clean, remove standing water and get **vaccinated**.

Key words

English	سنڌي	اردو
microorganism	خوردبيني جاندار	خوردبینی جاندار
microscope	خوردبين	خوردبین
bacteria	بيڪٽيريا	بیکٹیریا
virus	وائرس	وائرس
fungi	فنگس	فنجائی
yeast	خمير	خمیر
mould	ڦوڪڙ	پھپھوندی
antibiotic	اينٽي بايوٽڪ	اینٹی بایوٹک
vaccine	ويڪسين	ویکسین
infection	انفيڪشن	انفیکشن
hygiene	صفائي	حفظان صحت

Experiment: Does yeast make dough rise?



You need

- Two small bowls of flour
- Half a spoon of yeast
- One spoon of sugar
- Lukewarm water
- Two glasses and a marker

Steps

1. Mix the yeast and sugar into the first bowl of flour. Add nothing to the second bowl.
2. Add a little lukewarm water to both and knead two soft doughs.
3. Press each dough into its own glass and mark its height with the marker.

4. Cover both glasses and keep them in a warm place for 2 hours.
5. Compare the height of each dough with its mark.

Safety: Use lukewarm water, not hot water. Ask an adult to help, and wash your hands afterwards.

What did we learn? The dough with yeast rose much more. Yeast is a tiny fungus that feeds on sugar and gives out a gas. The gas makes small bubbles in the dough, so it rises. This is an example of useful microbes.

Practice questions

Circle the correct answer

1. **What do we need to see microorganisms?**
   - **(a)** A microscope
   - **(b)** A telescope
   - **(c)** A mirror
   - **(d)** Only our eyes
2. **Where are microorganisms found?**
   - **(a)** Only in dirty water
   - **(b)** Everywhere: in soil, air, water and our bodies
   - **(c)** Only in hospitals
   - **(d)** Only on plants
3. **Which are the smallest microbes?**
   - **(a)** Mushrooms
   - **(b)** Bacteria
   - **(c)** Viruses
   - **(d)** Yeast
4. **What is the green mould on old bread?**
   - **(a)** A fungus
   - **(b)** A virus
   - **(c)** A plant
   - **(d)** Soil
5. **Which microorganism makes dough rise?**
   - **(a)** Virus
   - **(b)** Yeast
   - **(c)** Mosquito
   - **(d)** Cholera bacteria
6. **Penicillin is made from what?**
   - **(a)** A fungus
   - **(b)** A rock
   - **(c)** Sea water
   - **(d)** A virus
7. **What do we call breaking down dead leaves into the soil?**
   - **(a)** Infection
   - **(b)** Decomposition
   - **(c)** Vaccination
   - **(d)** Rising

8. **What spreads the dengue virus?**

(a) Honeybee (b) Mosquito

(c) Dog (d) Cat

9. **What is the best way to stay safe from germs?**

(a) Washing hands with soap before eating (b) Leaving food uncovered

(c) Drinking unboiled pond water (d) Keeping rubbish in the house

Write true or false

10. **All microorganisms are harmful.** _____

11. **A virus can only grow inside the cells of a living thing.** _____

12. **Bacteria turn milk into yoghurt.** _____

13. **Cholera is caused by a virus.** _____

14. **Boiling water kills the harmful germs in it.** _____

Match the columns

15. **Match each microbe with its group.**

1. Yeast		(a) Virus
2. Polio germ		(b) Bacteria
3. Yoghurt germs		(c) Fungus

Match the columns

16. **Match each disease with its cause.**

1. Typhoid		(a) Virus
2. Measles		(b) Fungus
3. Athlete's foot		(c) Bacteria

Worksheet: Microorganisms

Name: _____ Date: _____

A. Fill in the blanks

Word box: Viruses · Yeast · fungus · infection · microscope

1. An instrument that makes tiny things look bigger is called a _____.
2. _____ are the smallest microbes.
3. _____ makes dough rise.
4. Penicillin is made from a _____.
5. A disease caused by microbes is called an _____.

B. Put ✓ for true and ✗ for false

1. ☐ Bacteria are made of one cell.
2. ☐ Mushrooms are a kind of virus.
3. ☐ Vaccines protect us from diseases.
4. ☐ Food left uncovered never gets germs.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Yeast	( ) Common cold
2. Bacteria	( ) Polio drops
3. Virus	( ) Naan and bread
4. Vaccine	( ) Yoghurt

D. Answer in short

1. Name the three groups of microorganisms.

_____

_____

2. Give two examples of useful microbes.

_____

_____

3. Write three ways to protect ourselves from germs.

_____

_____

E. Draw and label

Draw three boxes: bacteria, virus and fungus. In each box draw a picture and write one example.

CHAPTER 3

Seeds: Structure and Germination

What we will learn Learn the parts of a seed, how a bean seed and a maize seed differ, and what a seed needs to germinate.

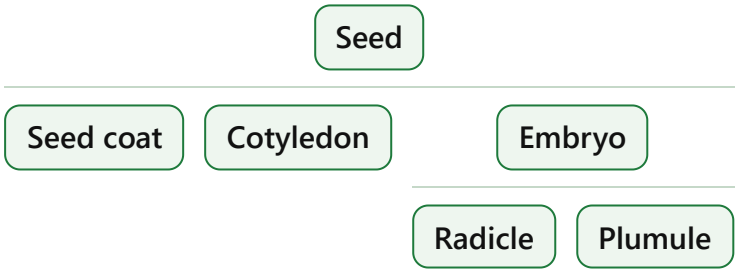
1 Parts of a seed

A new plant grows from a seed. Inside every seed there is a tiny plant and food for it.

- **Seed coat**: covers the seed and protects it. It has a tiny hole called the **micropyle**. Water goes in through it.
- **Embryo**: the baby plant inside the seed. The part that becomes the **root** is called the **radicle**, and the part that becomes the **stem and leaves** is called the **plumule**.
- **Cotyledons (seed leaves)**: they store **food** for the baby plant.

Seeds with two cotyledons are called **dicot seeds**, like beans, peas and chickpeas. Seeds with one cotyledon are called **monocot seeds**, like maize, wheat and rice.

Diagram: the parts and what they do



Seed (بچ · بیج)
A new plant grows from a seed. It holds a tiny plant and its food.

Seed coat (بچ جو کل · بیج کا چھلکا)
Protects the seed. Water enters through its tiny hole, the micropyle.

Cotyledon (بچ پن · دال پتا)
Stores food for the baby plant. A bean has two; maize has one.

Embryo (قوت · جنین)
The baby plant inside the seed, which grows into a new plant.

Radicle (ریڈیکل · ریڈیکل)
Comes out first and grows down to become the root.

Plumule (پلیومول · پلیومیول)
Grows up towards the light to become the stem and leaves.

2 A bean seed and a maize seed

- A **bean seed** is a dicot. Take the coat off a soaked bean and it opens easily into two halves. The two halves are the two **cotyledons**, and the tiny **embryo** is attached between them.
- A **maize grain** is a monocot. It has one cotyledon, an embryo and a large store of food called the **endosperm**.

In a bean, food is stored in the cotyledons. In maize, most of the food is stored in the endosperm.

We eat many seeds as food, such as lentils, chickpeas, peanuts and maize.

3 How does a seed germinate?

When a tiny plant comes out of a seed and starts to grow, we call it **germination**. The young new plant is called a **seedling**.

First the **radicle** comes out and grows down into the soil to become the root. Then the **plumule** grows up towards the light.

- **In a bean**: the stem bends like a hook, pushes out of the soil and pulls the cotyledons **above the soil**. The cotyledons turn green and make food for a few days, then dry up and fall off.
- **In maize**: the cotyledon stays **under the soil**. The soft plumule grows straight up inside a protective sheath.



Picture: Judgefloro · CC0 · Wikimedia Commons

4 What seeds need to germinate

Not every seed germinates. A seed needs three things to germinate:

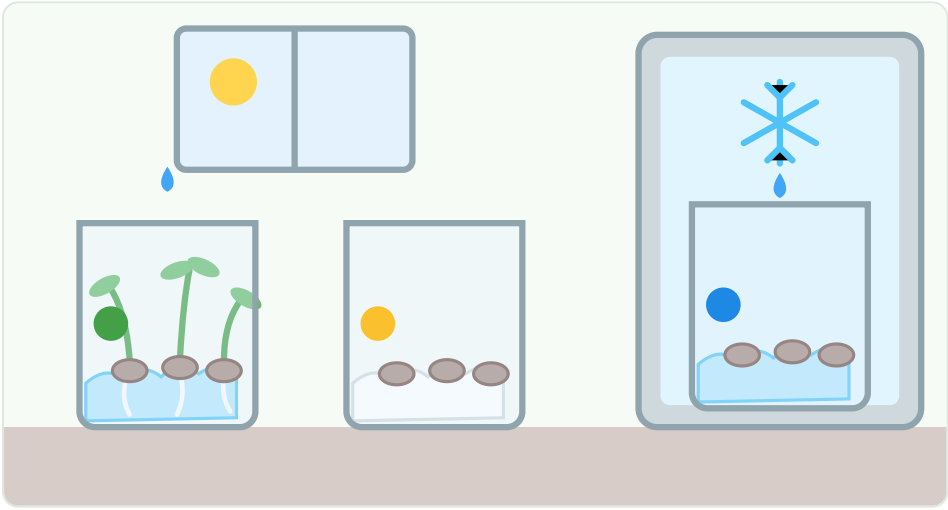
- **Water**: the seed soaks up water and swells, and its coat softens and splits. A dry seed does not germinate.
- **Air (oxygen)**: the seed has to breathe to get energy.
- **The right temperature**: most seeds germinate when it is warm. They do not germinate when it is very cold or very hot.

Some seeds also need light, while others germinate in the dark. A seed **does not need soil** to germinate: it can sprout on wet cotton. But to keep growing, the plant needs soil, sunlight and food.

Key words

English	سنڌي	اردو
seed	ٻج	بیج
seed coat	ٻج جو کل	بیج کا چھلکا
micropyle	مائڪروپائل	مائیکروپائل
embryo	ڏوٽ	جنین
radicle	ريڊيڪل	ریڈیکل
plumule	پليومول	پلیومیول
cotyledon	ٻج پن	دال پتا
endosperm	اينڈواسپرم	اینڈوسپرم
germination	ڦٽڻ	اگنا
seedling	ٻوٽڙو	پنیری

Experiment: What does a seed need to germinate?



You need

- Three small jars or cups
- Cotton wool
- 12 bean or chickpea seeds
- Water and labels

Steps

1. Label the jars A, B and C, and put cotton wool and 4 seeds in each.
2. Make the cotton in jar A wet and keep it in the room.
3. Leave the cotton in jar B dry and keep it in the room too.
4. Make the cotton in jar C wet and put it in the fridge.
5. Check every day for 5 days and keep A and C wet. Write down which seeds grew roots.

Safety: Ask an adult at home before using the fridge. Do not eat the seeds, and wash your hands afterwards.

What did we learn? Only the seeds in jar A germinated, where they had water, air and warmth. Jar B had no water and jar C was too cold. So a seed needs water, air and the right temperature to germinate.

Practice questions

Circle the correct answer

1. **Which part of a seed protects it?**

   (a) Seed coat (b) Radicle
   (c) Plumule (d) Endosperm

2. **Which part of the embryo becomes the root?**

   (a) Plumule (b) Radicle
   (c) Cotyledon (d) Micropyle

3. **Which of these is a dicot seed?**

   (a) Maize (b) Wheat
   (c) Bean (d) Rice

4. **Where is most of the food stored in a maize grain?**

   (a) In the endosperm (b) In the radicle
   (c) In the seed coat (d) In the micropyle

5. **How many cotyledons does a bean seed have?**

   (a) One (b) Two
   (c) Three (d) None

6. **What comes out of the seed first during germination?**

   (a) Leaves (b) A flower
   (c) The radicle (d) A fruit

7. **In which seed do the cotyledons come above the soil?**

   (a) Bean (b) Maize
   (c) Wheat (d) None of them

8. **Which of these does a seed NOT need to germinate?**

   (a) Water (b) Air
   (c) The right temperature (d) Soil

9. **Why does a seed need air (oxygen)?**

(a) To breathe and get energy (b) To change colour

(c) To become heavy (d) To dry out

Write true or false

10. **Cotyledons store food for the baby plant.** _____

11. **A maize grain has two cotyledons.** _____

12. **In maize, the cotyledon stays under the soil.** _____

13. **A dry seed can germinate without water.** _____

14. **A seed can germinate on wet cotton.** _____

Match the columns

15. Match each part of a seed with its job.

1. Radicle		(a) Becomes the stem and leaves
2. Plumule		(b) Lets water in
3. Micropyle		(c) Becomes the root

Match the columns

16. Match each seed with its type.

1. Pea		(a) Monocot seed
2. Wheat		(b) Dicot seed

Worksheet: Seeds: Structure and Germination

Name: _____ Date: _____

A. Fill in the blanks

Word box: endosperm · germination · plumule · seed coat · temperature

1. The cover on the outside of a seed is called the _____ .
2. The _____ of the embryo grows up into the stem and leaves.
3. In maize, food is stored in the _____ .
4. A tiny plant coming out of a seed and growing is called _____ .
5. A seed needs water, air and the right _____ to germinate.

B. Put ✓ for true and ✗ for false

1. ☐ A bean seed is a dicot seed.
2. ☐ The plumule comes out first during germination.
3. ☐ A seed must have soil to germinate.
4. ☐ A seed takes in water through the micropyle.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Radicle	( ) Monocot seed
2. Plumule	( ) Dicot seed
3. Maize	( ) Root
4. Chickpea	( ) Stem and leaves

D. Answer in short

1. Name the three main parts of a seed.

2. Write one difference between a bean seed and a maize grain.

3. What three things does a seed need to germinate?

E. Draw and label

Draw an opened bean seed and label the seed coat, cotyledons, radicle and plumule.

CHAPTER 4

Environmental Pollution

What we will learn Learn what pollution is, how air, water and land get polluted, and how we can cut waste and keep our surroundings clean.

1 What is pollution?

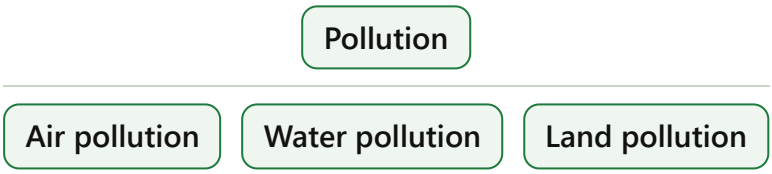
When things that are **harmful** to people, animals and plants get mixed into air, water or land, we call it **pollution**.

The harmful things that cause pollution are called **pollutants**, such as smoke, dirty water, plastic bags and chemicals from factories.

Most pollution is caused by the things **people** do.

There are three main kinds of pollution: **air pollution**, **water pollution** and **land pollution**.

Diagram: the parts and what they do



Pollution (آلودگي · آلودگی)
Harmful things getting mixed into air, water or land.

Air pollution (هوا جي آلودگي · ہوائی آلودگی)
Smoke from vehicles, factories and burning rubbish. It causes coughing and asthma.

Water pollution (پاڻيءَ جي آلودگي · آبی آلودگی)
Sewage, chemicals and rubbish in water. It causes cholera and typhoid.

Land pollution (زمين جي آلودگي · زمینی آلودگی)
Throwing plastic, bottles and rubbish on the ground. It spoils soil and underground water.

2 Air pollution

Causes of air pollution:

- **Smoke** from cars, buses and rickshaws.
- Smoke and gases from **factory** chimneys.
- **Burning rubbish** and burning fuels such as wood, coal and oil.

Breathing polluted air causes **coughing**, **asthma**, burning eyes, headaches and lung diseases. In big cities like Karachi, smoke from vehicles makes air pollution very high.



Picture: Adina Voicu · CC0 · Wikimedia Commons

3 Water and land pollution

Causes of **water pollution**: letting dirty water from homes (**sewage**) into rivers or the sea without cleaning it, chemicals from factories, and pesticides and fertilisers washing off farms.

Drinking polluted water causes **cholera**, **typhoid** and diarrhoea. Fish and other living things also die in polluted water.

Land pollution happens when rubbish such as plastic bags, bottles, paper and tin cans is thrown on the ground.

Heaps of rubbish spoil the soil and the water under the ground. Animals get sick from eating plastic, and flies and mosquitoes breed in rubbish.

4 Biodegradable and non-biodegradable waste

- **Biodegradable waste**: rots away into the soil after some time because of microbes. Examples: fruit and vegetable peels, leftover food, dry leaves and paper.
- **Non-biodegradable waste**: does not rot even in a hundred years. Examples: **plastic**, glass, tin cans and polythene.

Plastic that reaches rivers and the sea harms fish, turtles and birds.

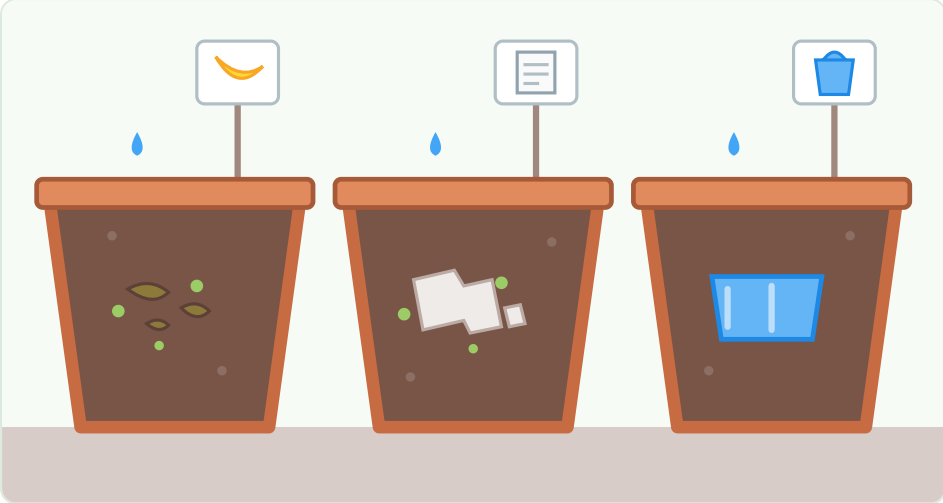
We can reduce pollution: **reduce**, **reuse** and **recycle**. Use cloth bags, do not burn rubbish, put rubbish in the bin, walk or share rides, and **plant trees**.

Key words

English	سنڌي	اردو
pollution	آلودگي	آلودگی
pollutant	آلوده ڪندڙ	آلودہ کار
environment	ماحول	ماحول

English	سنڌي	اردو
smoke	دونهون	دھواں
sewage	سيوريج	سیوریج
waste	ڪچرو	کچرا
biodegradable	گرندڙ ڪچرو	گلنے والا کچرا
non-biodegradable	نه گرندڙ ڪچرو	نہ گلنے والا کچرا
recycle	ري سائيڪل	ری سائیکل
asthma	دمو	دمہ

Experiment: Which waste rots away in soil?



You need

- Three small pots filled with soil
- A banana or vegetable peel
- A piece of paper
- A piece of plastic bag
- Water and labels

Steps

1. Bury one item about 5 centimetres deep in each pot.
2. Label each pot with the name of the item.
3. Sprinkle a little water every other day to keep the soil damp.
4. After three weeks, dig carefully and look at all three items.
5. Write down which item rotted and which stayed the same.

Safety: Wash your hands with soap after touching soil and waste. Do not use glass or sharp things.

What did we learn? The peel rotted a lot and the paper became soft and started to break up, but the plastic stayed exactly the same. Microbes in the soil break down biodegradable waste but cannot break down plastic. So we should use less plastic.

Practice questions

Circle the correct answer

1. **What is pollution?**

   (a) Harmful things getting into air, water or land
   (b) Planting trees
   (c) Rain falling
   (d) Putting rubbish in a bin

2. **What are the harmful things that cause pollution called?**

   (a) Manure
   (b) Pollutants
   (c) Biodegradables
   (d) Vaccines

3. **Which of these causes air pollution?**

   (a) Riding a bicycle
   (b) Planting trees
   (c) Burning rubbish
   (d) Walking

4. **Which illness can breathing polluted air cause?**

   (a) Asthma
   (b) Tooth decay
   (c) A broken bone
   (d) Athlete's foot

5. **What is dirty water from homes called?**

   (a) Rainwater
   (b) Sewage
   (c) Steam
   (d) Ice

6. **Which disease can come from drinking polluted water?**

   (a) Cholera
   (b) Weak eyesight
   (c) Asthma
   (d) None

7. **Which of these is biodegradable?**

   (a) Plastic bottle
   (b) Glass
   (c) Banana peel
   (d) Tin can

8. **What should we use instead of a plastic bag?**

   (a) Another plastic bag
   (b) A cloth bag
   (c) Polythene
   (d) Nothing, just burn it

9. **What does recycling mean?**

   (a) Making new things from old things
   (b) Burning rubbish
   (c) Throwing rubbish in a river
   (d) Buying more plastic

Write true or false

10. **Most pollution is caused by people.** _____

11. **Burning rubbish makes the air clean.** _____

12. **Pesticides washing off farms can pollute water.** _____

13. **Plastic rots into the soil in a few days.** _____

14. **Planting trees helps keep the air clean.** _____

Match the columns

15. **Match each cause with the kind of pollution.**

1. Smoke from cars		(a) Water pollution
2. Sewage in the sea		(b) Land pollution
3. Heaps of rubbish		(c) Air pollution

Match the columns

16. **Match each piece of waste with its kind.**

1. Dry leaves		(a) Non-biodegradable
2. Plastic bag		(b) Biodegradable

Worksheet: Environmental Pollution

Name: _____ Date: _____

A. Fill in the blanks

Word box: Plastic · asthma · pollutants · recycling · sewage

1. The harmful things that cause pollution are called _____ .
2. Dirty water from homes is called _____ .
3. Polluted air can cause _____ .
4. _____ does not rot even in a hundred years.
5. Making new things from old things is called _____ .

B. Put ✓ for true and ✗ for false

1. ☐ Smoke from factories pollutes the air.
2. ☐ Fish live happily in polluted water.
3. ☐ Fruit peels are biodegradable.
4. ☐ Burning rubbish is a good way to reduce pollution.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Air pollution	( ) Plastic bags
2. Water pollution	( ) Vegetable peels
3. Land pollution	( ) Smoke from vehicles
4. Biodegradable waste	( ) Factory chemicals

D. Answer in short

1. Name the three kinds of pollution.

_____

_____

2. What is the difference between biodegradable and non-biodegradable waste?

_____

_____

3. Write three ways to reduce pollution.

_____

_____

E. Draw and label

Make a poster to keep your school clean. Draw a rubbish bin on it and write a slogan.

CHAPTER 5

Matter and Its Changes

What we will learn Learn what matter is, its three states, and the difference between physical and chemical changes.

1 What is matter?

Anything that **takes up space** and has **mass (weight)** is called **matter**.

Books, water, air, stones and our own bodies — all are matter.

We cannot see air, but it is matter too. When you blow into a balloon it gets bigger, because air takes up space.

Matter is made of tiny **particles** that are too small to see with our eyes.

2 The three states of matter

Matter is found in three states:

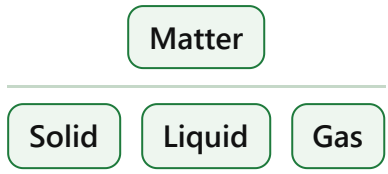
- **Solid**: has a fixed shape and a fixed size. Its particles are packed very close together. Examples: stone, wood, ice.
- **Liquid**: has a fixed size but no fixed shape. It takes the shape of the container it is poured into. Examples: water, milk, oil.
- **Gas**: has no fixed shape and no fixed size. It spreads out to fill all the space. Examples: air, steam, oxygen.

Heating or cooling can change matter from one state to another.



Picture: Stephanie Costa, Sven Hoppe et al. · CC BY-SA 3.0 · Wikimedia Commons

Diagram: the parts and what they do



Matter (مادو · ماده)
Anything that takes up space and has mass.

Solid (ٺوس · ٹھوس)
Fixed shape and size. Particles packed close. Examples: stone, wood, ice.

Liquid (مائع · مائع)
Fixed size but no fixed shape. Takes the shape of its container. Examples: water, milk, oil.

Gas (گئس · گیس)
No fixed shape or size. Spreads to fill all the space. Examples: air, steam.

3 Physical and chemical changes

Matter can go through two kinds of change:

- **Physical change**: no new substance is made. Often the change can be reversed. Examples: ice melting, water turning into vapour, paper tearing, sugar dissolving in water.
- **Chemical change**: a **new substance** is made and the change cannot easily be reversed. Examples: paper burning, an egg cooking, iron getting **rust**, milk turning into yoghurt.

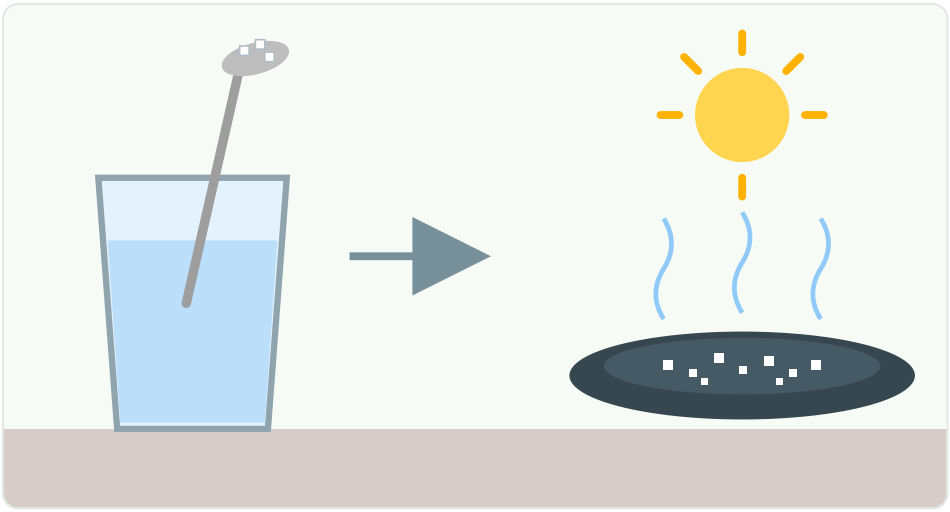
In a chemical change the colour often changes, heat or light is given out, or a new smell appears.

Key words

English	سنڌي	اردو
matter	مادو	ماده
mass	ڪميت	کمیت
particles	ذرڙا	ذرات
solid	ٺوس	ٹھوس
liquid	مائع	مائع
gas	گئس	گیس
physical change	طبعي تبديلي	طبعی تبدیلی
chemical change	ڪيميائي تبديلي	کیمیائی تبدیلی
dissolve	ڳرڻ (حل ٿيڻ)	گھلنا

English	سنڌي	اردو
rust	ڪٽ	زنگ

Experiment: Can we get dissolved salt back?



You need

- One spoon of salt
- Half a glass of water
- A dark or deep plate
- A spoon

Steps

1. Put the salt in the water and stir. The salt will disappear from sight.
2. Pour a little of the salty water into the plate.
3. Keep the plate in the sun for 2 or 3 days.
4. When all the water has dried up, look at the plate.

Safety: Do not drink the salty water, and wash your hands after the experiment.

What did we learn? White grains of salt are left on the plate. The water evaporated but the salt stayed the same. Salt dissolving in water is a physical change, because no new substance was made and we got the salt back.

Practice questions

Circle the correct answer

1. **What is matter?**

   (a) Anything that takes up space and has mass

   (b) Only things we can see

   (c) Only living things

   (d) Only solid things

2. **Why does a balloon get bigger when we blow into it?**

   (a) Air takes up space

   (b) The balloon is alive

   (c) The balloon melts

   (d) Air has no mass

3. **Which state has both a fixed shape and a fixed size?**

(a) Solid **(b)** Liquid

(c) Gas **(d)** None

4. **Milk takes the shape of the glass it is poured into. Which state is milk in?**

(a) Solid **(b)** Liquid

(c) Gas **(d)** Particle

5. **Which of these is a gas?**

(a) Wood **(b)** Oil

(c) Oxygen **(d)** Stone

6. **Which of these is a chemical change?**

(a) Ice melting **(b)** Paper tearing

(c) Paper burning **(d)** Sugar dissolving in water

7. **Which of these is a physical change?**

(a) An egg cooking **(b)** Iron rusting

(c) Water freezing into ice **(d)** Wood burning

8. **What happens in a chemical change?**

(a) A new substance is made **(b)** Only the shape changes

(c) Nothing happens **(d)** The thing always goes back to how it was

9. **Matter is made of tiny what?**

(a) Particles **(b)** Leaves

(c) Drops **(d)** Seeds

Write true or false

10. **Air is not matter because we cannot see it.** _____

11. **A gas spreads out to fill all the space.** _____

12. **In a solid, the particles are very far apart.** _____

13. **A cooked egg cannot be turned back into a raw egg.** _____

14. **Iron rusting is a physical change.** _____

Match the columns

15. **Match each thing with its state.**

1. Wood		(a) Liquid
2. Oil		(b) Gas
3. Steam		(c) Solid

Match the columns

16. **Match each change with its kind.**

1. Paper tearing		(a) Chemical change
2. Milk turning into yoghurt		(b) Physical change

Worksheet: Matter and Its Changes

Name: _____ Date: _____

A. Fill in the blanks

Word box: container · matter · new · rust · solid

1. Anything that takes up space and has mass is called _____ .
2. A _____ has a fixed shape and a fixed size.
3. A liquid takes the shape of its _____ .
4. In a chemical change a _____ substance is made.
5. Damp air makes _____ form on iron.

B. Put ✓ for true and ✗ for false

1. ☐ Our body is also matter.
2. ☐ A gas has a fixed shape.
3. ☐ Paper burning is a chemical change.
4. ☐ Melting ice can never be reversed.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Stone	( ) Gas
2. Milk	( ) Chemical change
3. Air	( ) Solid
4. Egg cooking	( ) Liquid

D. Answer in short

1. Write the three states of matter with one example of each.

2. Write one difference between a physical and a chemical change.

3. How do we know that air takes up space?

E. Draw and label

Draw three boxes: solid, liquid and gas. In each box draw one thing and write its name.

CHAPTER 6

Force and Simple Machines

What we will learn Learn about pushes and pulls, what forces do, friction, and six simple machines that make work easier.

1 What is a force?

A **push** or a **pull** is called a **force**. To open a door we push it or pull it.

A force can:

- **start** something moving, like kicking a football.
- **stop** something that is moving, like catching a ball.
- change the **direction** or **speed** of something, like hitting a cricket ball with a bat.
- change the **shape** of something, like making a cup out of clay.

2 Some kinds of force

- **Muscular force**: the strength of our body, like lifting a bag or riding a bicycle.
- **Gravity**: the Earth pulls everything down towards it. That is why a ball thrown up comes back down.
- **Friction**: when two surfaces rub together, friction slows a moving thing down. There is more friction on rough surfaces and less on smooth ones. Friction is also what lets us walk without slipping.
- **Magnetic force**: a magnet pulls iron towards it, even without touching it.

3 Simple machines

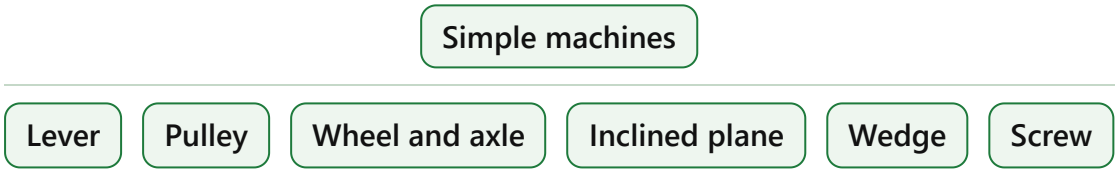
Simple machines are tools that make work **easier** using less force. There are six simple machines:

- **Lever**: a strong bar that turns on one point. Examples: a see-saw, a spoon used to open a tin lid.
- **Pulley**: a wheel with a rope running over it. Examples: pulling water from a well, raising a flag.
- **Wheel and axle**: a wheel that turns together with a rod (axle). Examples: a bicycle, a cart, a door handle.
- **Inclined plane (ramp)**: a sloping surface that makes it easier to move heavy things up. Example: a plank for loading a truck.
- **Wedge**: used to cut or split things apart. Examples: an axe, a knife.
- **Screw**: holds things together when it is turned. Examples: a bottle cap, a screw.



Picture: Killarnee · CC BY-SA 4.0 · Wikimedia Commons

Diagram: the parts and what they do



Simple machines (سادا مشينون · سادہ مشینیں)
Tools that make work easier with less force.

Lever (ليور · لیور)
A bar that turns on one point. Examples: see-saw, opening a lid with a spoon.

Pulley (پُلي · پُلی)
A wheel and rope that make lifting easier. Examples: a well, a flagpole.

Wheel and axle (ڦيٽو ۽ ڏُرو · پہیہ اور دُھرا)
A wheel and rod that turn together. Examples: bicycle, cart.

Inclined plane (لاهي واري سطح · ڈھلوان سطح)
A slope that makes moving heavy things up easier. Example: a ramp.

Wedge (ويج · فانہ)
A sharp edge that splits things. Examples: axe, knife.

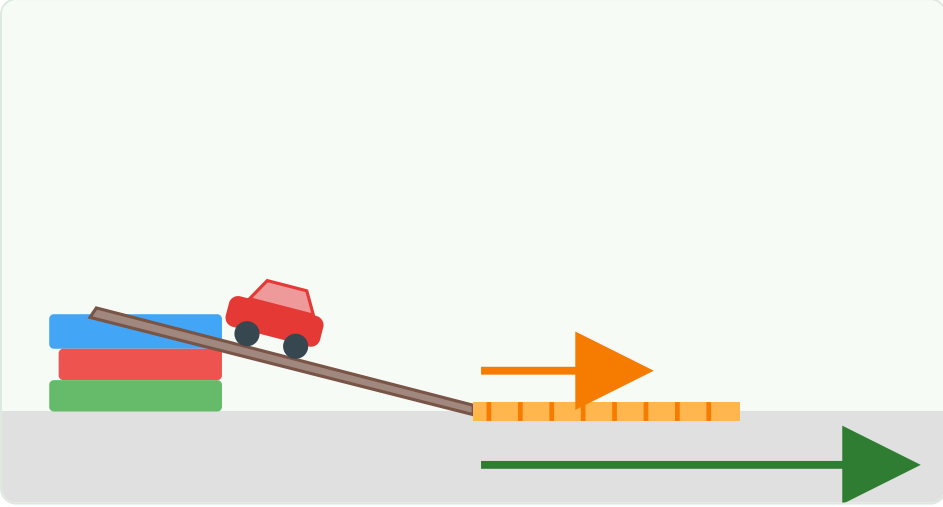
Screw (پيچ · پیچ)
Holds things tightly when turned. Example: a bottle cap.

Key words

English	سنڌي	اردو
force	قوت	قوت
push	ڌڪو	دھکا

English	سنڌي	اردو
pull	ڇڪ	کھنچاؤ
gravity	ڪشش ثقل	کششِ ثقل
friction	رگڑ	رگڑ
simple machine	سادي مشين	سادہ مشین
lever	ليور	لیور
pulley	پُلي	پُلی
wheel and axle	ڦيٽو ۽ ڏُرو	پہیہ اور دُھرا
inclined plane	لاهي واري سطح	ڈھلوان سطح

Experiment: Friction on rough and smooth surfaces



You need

- A toy car or a small ball
- A book and a board or thick card
- A towel or a rug

Steps

1. Lean the board on the book to make a ramp.
2. Let the car roll down and see how far it goes on the smooth floor.
3. Now spread the towel at the bottom of the ramp and let the car roll again.
4. Compare how far the car went each time.

Safety: Do the experiment on the floor and never throw the car at anyone.

What did we learn? The car stops sooner on the towel. The rough towel makes more friction, which slows the car down. The smooth floor has less friction, so the car goes further.

Practice questions

Circle the correct answer

1. **What is a force?**

   (a) A push or a pull **(b)** A colour

   (c) A sound **(d)** A plant

2. **When we make a cup from clay, what does the force change?**

   (a) The colour **(b)** The shape

   (c) The mass **(d)** The taste

3. **Why does a ball thrown up come back down?**

   (a) Because of friction **(b)** Because of gravity

   (c) Because of the wind **(d)** Because of light

4. **On which surface is there the most friction?**

   (a) Ice **(b)** Smooth glass

   (c) A rough stony road **(d)** An oily floor

5. **A see-saw is which simple machine?**

   (a) Lever **(b)** Screw

   (c) Wedge **(d)** Pulley

6. **Which simple machine is used to pull water from a well?**

   (a) Wedge **(b)** Pulley

   (c) Inclined plane **(d)** Screw

7. **An axe is which simple machine?**

   (a) Wheel and axle **(b)** Pulley

   (c) Wedge **(d)** Lever

8. **A plank used to load heavy things onto a truck is which simple machine?**

   (a) Inclined plane **(b)** Screw

   (c) Pulley **(d)** Wedge

9. **Which force do we use to lift a school bag?**

   (a) Magnetic force **(b)** Muscular force

   (c) Friction **(d)** Light

Write true or false

10. **A force can stop a moving object.** _____

11. **Friction makes moving things go faster.** _____

12. **A magnet can pull iron without touching it.** _____

13. **Simple machines make work harder.** _____

14. **A bicycle has a wheel and axle.** _____

Match the columns

15. Match each simple machine with an example.

1. Lever		(a) Raising a flag
2. Pulley		(b) Knife
3. Wedge		(c) Bottle cap
4. Screw		(d) See-saw

Match the columns

16. Match each action with what the force does.

1. Kicking a football		(a) Stops it
2. Catching a ball		(b) Changes its shape
3. Making a clay cup		(c) Starts it moving

Worksheet: Force and Simple Machines

Name: _____ Date: _____

A. Fill in the blanks

Word box: force · friction · gravity · lever · pulley

1. A push or a pull is called a _____.
2. The Earth pulls everything down because of _____.
3. There is more _____ on a rough surface.
4. A _____ is used to pull water from a well.
5. A see-saw is a _____.

B. Put ✓ for true and ✗ for false

1. ☐ A force can change the direction of an object.
2. ☐ There is a lot of friction on ice.
3. ☐ A knife is a wedge.
4. ☐ Simple machines need more force to do work.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Wheel and axle	( ) Axe
2. Screw	( ) Ramp
3. Wedge	( ) Cart
4. Inclined plane	( ) Bottle cap

D. Answer in short

1. Write two things a force can do.

2. How does friction help us?

3. Name the six simple machines.

E. Draw and label

Draw two simple machines used at your home or school and write their names.

CHAPTER 7

Light and Sound

What we will learn Learn about sources of light, how shadows form, transparent and opaque things, and how sound is made and travels.

1 Sources of light

Things that give out their own **light** are called **luminous objects** or **sources** of light. Examples: the sun, a bulb, a candle, a firefly.

The sun is a **natural** source of light. Bulbs and torches are **man-made** sources.

The moon, a book and a chair do not give out their own light. They are **non-luminous**. The moon reflects sunlight, so it looks bright.

We see an object when light from it enters our **eyes**.



Picture: Kelvinsong · CC0 · Wikimedia Commons

2 Light and shadows

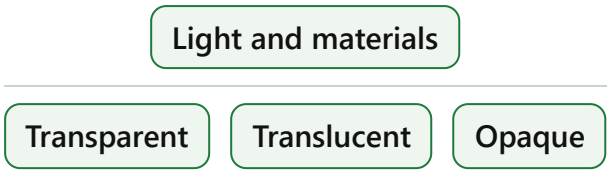
Light travels in **straight lines**.

When an object blocks the path of light, a dark area forms behind it. This is called a **shadow**.

Materials are of three kinds:

- **Transparent**: all the light passes through and we can see clearly. Examples: clear glass, clean water.
- **Translucent**: some light passes through and things look blurry. Examples: butter paper, frosted glass.
- **Opaque**: no light passes through and a shadow forms. Examples: wood, a book, a wall.

Diagram: the parts and what they do



Light and materials (روشني ۽ شيون · روشنی اور اشیاء)
Light passes through some materials but not others.

Transparent (شفاف · شفاف)
All light passes through. Examples: clear glass, clean water.

Translucent (نيم شفاف · نیم شفاف)
Some light passes through; things look blurry. Example: butter paper.

Opaque (غير شفاف · غیر شفاف)
No light passes through and a shadow forms. Examples: wood, a book.

3 Sound

Sound is made when something **vibrates** (moves quickly back and forth). Beat a drum or pluck a stretched rubber band and you can feel the vibration.

Sound can travel through **solids, liquids and gases**. It passes through air, water and even walls.

Sound cannot travel through **empty space**, because there is no matter there.

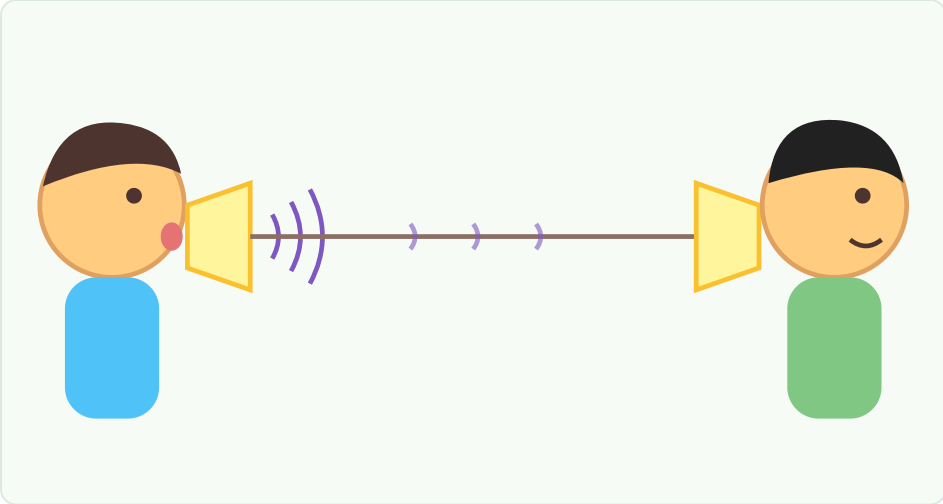
Some sounds are **loud** and some are **soft**. Very loud sounds can harm our ears, so we should protect our ears.

Key words

English	سنڌي	اردو
light	روشني	روشنی
luminous object	روشن شيءِ	روشن جسم
non-luminous object	غير روشن شيءِ	غیر روشن جسم
shadow	پاڇو	سایہ
transparent	شفاف	شفاف
translucent	نيم شفاف	نیم شفاف
opaque	غير شفاف	غیر شفاف
sound	آواز	آواز
vibration	ٿرٿراهٽ	ارتعاش

English	سنڌي	اردو
space (vacuum)	خلا	خلا

Experiment: Make a string telephone



You need

- Two paper or plastic cups
- About 5 metres of strong string
- A pencil (to make holes)

Steps

1. With an adult's help, make a small hole in the bottom of each cup with the pencil.
2. Push each end of the string through a hole and tie a knot inside the cup.
3. Stand far from a friend so the string is tight.
4. Speak softly into one cup while your friend listens with the other cup to their ear.

Safety: Ask an adult to help make the holes, and never shout into the cup.

What did we learn? Your friend can hear you clearly. Your voice makes the cup and the tight string vibrate, and the vibrations travel along the string (a solid) to the other cup. This shows that sound travels through solids.

Practice questions

Circle the correct answer

1. **Which of these gives out its own light?**

   (a) The moon
   (b) The sun
   (c) A mirror
   (d) A book

2. **Why does the moon look bright at night?**

(a) It reflects sunlight (b) It is on fire

(c) It has bulbs on it (d) It is a star

3. **When can we see an object?**

(a) When light from it enters our eyes (b) Only when it makes a loud sound

(c) When our eyes are closed (d) Only in the dark

4. **How does light travel?**

(a) In straight lines (b) In circles

(c) In wavy curves only (d) It does not travel

5. **When does a shadow form?**

(a) When an opaque object blocks the light (b) When there is no light at all

(c) When the object is transparent (d) When there is a sound

6. **Clear glass is which kind of material?**

(a) Transparent (b) Translucent

(c) Opaque (d) Luminous

7. **How is sound made?**

(a) By vibrations (b) By light

(c) By shadows (d) By cold

8. **Where can sound NOT travel?**

(a) In air (b) In water

(c) In empty space (d) Through a wall

9. **Which of these is a man-made source of light?**

(a) The sun (b) A firefly

(c) A torch (d) A star

Write true or false

10. **The moon gives out its own light.** _____

11. **Wood is opaque, so it makes a shadow.** _____

12. **Butter paper is translucent.** _____

13. **Sound cannot travel through water.** _____

14. **Very loud sounds can harm our ears.** _____

Match the columns

15. **Match each material with its kind.**

1. Clean water		(a) Translucent
2. Frosted glass		(b) Opaque
3. Wall		(c) Transparent

Match the columns

16. **Match each word with its meaning.**

1. Luminous object		(a) Dark area behind blocked light
2. Shadow		(b) Quick back-and-forth movement
3. Vibration		(c) Gives its own light

Worksheet: Light and Sound

Name: _____ Date: _____

A. Fill in the blanks

Word box: luminous · shadow · space · straight · vibrations

1. Things that give out their own light are called _____ objects.
2. Light travels in _____ lines.
3. A _____ forms behind an opaque object.
4. Sound is made by _____ .
5. Sound cannot travel through empty _____ .

B. Put ✓ for true and ✗ for false

1. ☐ The sun is a natural source of light.
2. ☐ Light cannot pass through clear glass.
3. ☐ Sound travels through air.
4. ☐ A book is a luminous object.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Candle	( ) Sound from vibration
2. Chair	( ) Translucent
3. Drum	( ) Luminous object
4. Butter paper	( ) Non-luminous object

D. Answer in short

1. Write three sources of light.

_____

_____

2. How is a shadow formed?

_____

_____

3. Through which three kinds of matter can sound travel?

_____

_____

E. Draw and label

Draw the sun, a tree and the tree's shadow. Show the path of light with arrows.

CHAPTER 8

Electricity and Magnets

What we will learn Learn about uses of electricity and safety, a simple circuit, conductors and insulators, and the poles of a magnet.

1 Electricity and safety

Electricity is a kind of energy. It lights bulbs, runs fans, cools the fridge and charges mobile phones.

We get electricity from **power stations** through wires. Small things like torches and clocks run on a **battery (cell)**.

Electricity is useful but it can also be **dangerous**:

- Never touch a switch or socket with wet hands.
- Never push anything into a socket.
- Do not fly kites near electric wires and poles.
- If you see a broken wire, do not touch it — tell an adult.

2 A simple circuit

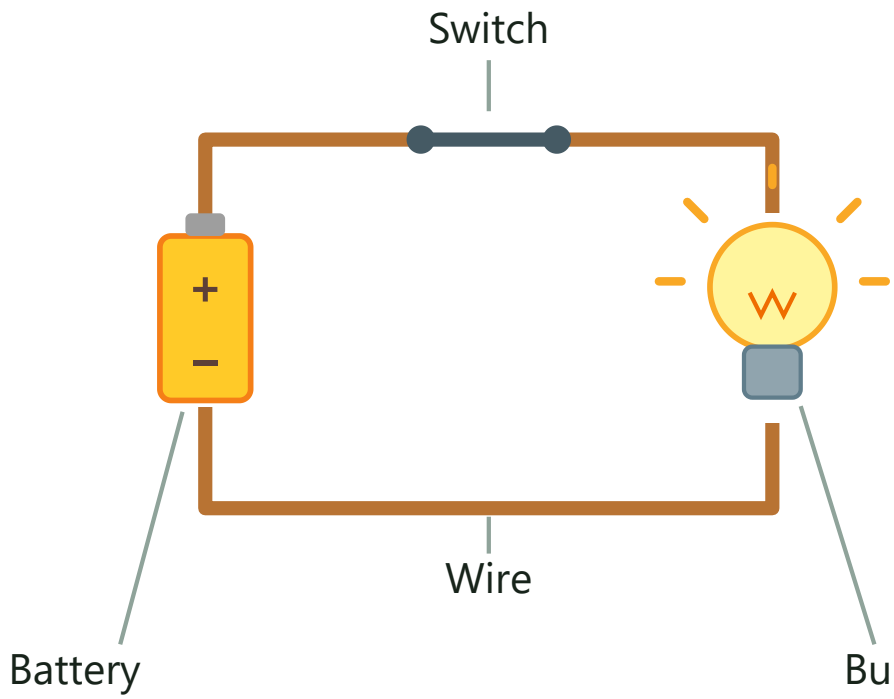
The path that electricity flows along is called a **circuit**. A simple circuit has:

- a **battery**, which pushes the electricity,
- **wires**, which carry the electricity from one place to another,
- a **switch**, which opens or closes the circuit,
- a **bulb**, which changes electricity into light.

When the path is complete (a **closed circuit**), the bulb lights up. When the path is broken somewhere (an **open circuit**), the bulb does not light.

Materials that let electricity pass are called **conductors**, like copper, iron and aluminium. Materials that do not let it pass are called **insulators**, like plastic, rubber, wood and glass. That is why wires are covered in plastic.

Diagram: the parts and what they do



Battery (بیټري · بیٹری)
Gives electrical energy and pushes electricity around the circuit.

Wire (تار · تار)
Copper wire carries electricity. Copper is a good conductor.

Switch (سوئچ · سوئچ)
Opens or closes the circuit. When closed, electricity flows and the bulb lights.

Bulb (بلب · بلب)
Changes electricity into light. It glows only in a complete circuit.

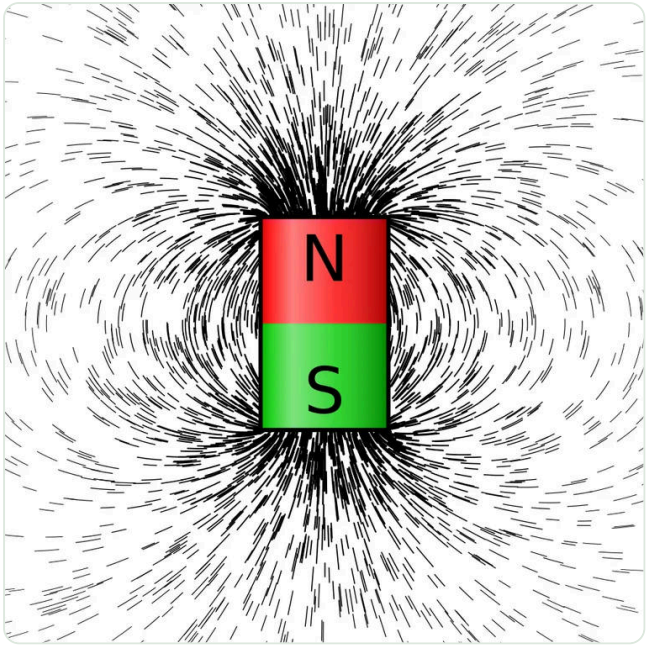
3 Magnets

A **magnet** pulls things made of **iron, steel and nickel** towards it. It does not pull paper, plastic or wood.

Every magnet has two **poles**: a **north pole (N)** and a **south pole (S)**. The pull is strongest at the ends of the magnet.

- **Unlike poles** (N and S) **attract** each other.
- **Like poles** (N and N, or S and S) **push each other away** (repel).

A magnet that hangs freely always points north–south. A **compass** works this way and helps us find our way.



Picture: Geek3 · CC BY-SA 4.0 · Wikimedia Commons

Key words

English	سنڌي	اردو
electricity	بجلي	بجلی
battery (cell)	بيٽري (سيل)	بیٹری (سیل)
circuit	سرڪٽ	سرکٹ
switch	سوئچ	سوئچ
conductor	موصل	موصل
insulator	غير موصل	غیر موصل
magnet	مقناطيس	مقناطیس
pole	قطب	قطب
attract	چڪڻ	کشش
repel	پري ڌڪڻ	دفع کرنا
compass	قطب نما	قطب نما

Experiment: What does a magnet pull?



You need

- A magnet (a fridge magnet will do)
- An iron nail, a pin, a steel spoon
- Paper, a plastic lid, an eraser, a wooden pencil

Steps

1. Put all the things on a table.
2. Guess which things will stick to the magnet and write your guesses in your notebook.
3. Bring the magnet close to each thing.
4. Make two lists: pulled and not pulled. Compare them with your guesses.

Safety: Keep magnets away from mobile phones, watches and computers. Handle the pin carefully.

What did we learn? The nail, pin and steel spoon stuck to the magnet because they are made of iron or steel. Paper, plastic, rubber and wood were not pulled. A magnet pulls only some metals.

Practice questions

Circle the correct answer

1. **What makes a torch work?**

   (a) A battery
   (b) Water
   (c) A magnet
   (d) Paper

2. **Which of these is safe?**

   (a) Pressing a switch with wet hands
   (b) Pushing a pin into a socket
   (c) Flying a kite in an open field away from wires
   (d) Touching a broken wire

3. **When does the bulb in a circuit light up?**

(a) When the circuit is complete (closed) (b) When a wire is broken

(c) When there is no battery (d) When the switch is open

4. **Which of these is a conductor?**

(a) Plastic (b) Rubber

(c) Copper (d) Wood

5. **Why are electric wires covered in plastic?**

(a) Plastic is an insulator and keeps us safe (b) To make the wire look pretty

(c) To make electricity faster (d) To make the wire heavy

6. **What does a magnet pull?**

(a) An iron nail (b) Paper

(c) A plastic spoon (d) A wooden pencil

7. **What happens when the N poles of two magnets are brought close?**

(a) They push each other away (b) They stick together

(c) They melt (d) Nothing at all

8. **Which tool uses a magnet to help us find directions?**

(a) A compass (b) A thermometer

(c) A weighing scale (d) A watch

9. **What does a switch do in a circuit?**

(a) Opens or closes the circuit (b) Makes electricity

(c) Gives light (d) Makes a magnet

Write true or false

10. **Rubber is an insulator.** _____

11. **A bulb lights up in an open circuit.** _____

12. **A magnet has two poles.** _____

13. **A magnet pulls paper.** _____

14. **Touching a socket with wet hands is dangerous.** _____

Match the columns

15. **Match each part of a circuit with its job.**

1. Battery		(a) Carries the electricity
2. Wire		(b) Opens or closes the circuit
3. Switch		(c) Gives light

4. Bulb		(d) Pushes the electricity

Match the columns

16. Match each pair of poles with what happens.

1. N and S		(a) Push each other away
2. S and S		(b) Attract each other

Worksheet: Electricity and Magnets

Name: _____ Date: _____

A. Fill in the blanks

Word box: battery · compass · conductors · poles · switch

1. A torch runs on a _____ .
2. Materials that let electricity pass are called _____ .
3. A _____ opens or closes a circuit.
4. A magnet has two _____ .
5. A _____ helps us find directions.

B. Put ✓ for true and ✗ for false

1. ☐ Electricity passes easily through plastic.
2. ☐ A bulb lights up in a closed circuit.
3. ☐ Unlike poles attract each other.
4. ☐ It is safe to fly a kite near electric wires.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Copper	( ) Gives light
2. Rubber	( ) Push away
3. Bulb	( ) Conductor
4. N and N poles	( ) Insulator

D. Answer in short

1. Write two ways to stay safe with electricity.

_____

_____

2. Name the four parts of a simple circuit.

_____

_____

3. Write two uses of magnets.

_____

_____

E. Draw and label

Draw a simple circuit with a battery, wires, a switch and a bulb, and label each part.

CHAPTER 9

Soils

What we will learn Learn what soil is made of, its properties and types, and which living things live in it.

1 What is soil made of?

The soft top layer on the surface of the Earth is called **soil**. Plants grow in soil.

Soil is made from **rocks**. Heat, cold, rain and wind break rocks into tiny pieces over thousands of years.

Soil contains:

- **Mineral particles**: grains of sand, silt and clay.
- **Humus**: rotted leaves, plants and dead animals. Humus makes soil rich.
- **Air** and **water**, and many **living things**.



Picture: USDA NRCS Montana · Public domain · Wikimedia Commons

2 Properties of soil

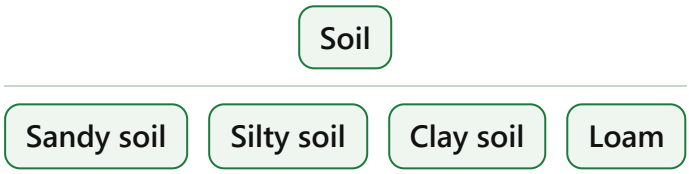
- **Colour**: soil can be red, brown, black or grey. Black and dark brown soils usually contain a lot of humus.
- **Texture**: rub some wet soil between your fingers. Sand feels **gritty**, silt feels **smooth and silky**, and clay feels **sticky**.
- **Compaction**: in tightly packed, hard soil, little air and water can get in and roots grow with difficulty.
- **Holding water and drainage**: some soils let water run through quickly, while others hold it for a long time. Good soil holds enough water and lets the extra water drain away.

3 Types of soil

- **Sandy soil**: large, gritty grains. Water drains through it very fast, so it holds little water and few nutrients. It is found in deserts like the Thar.
- **Silty soil**: fine, smooth grains. It holds water well and is fertile. It is found along the banks of the River Indus.
- **Clay soil**: very tiny grains. It is sticky when wet and hard when dry. It holds a lot of water, so water stands on it.
- **Loam**: a mixture of sand, silt, clay and humus. It holds the right amount of air and water, so it is the **best soil for plants**.

Grain size: sand is the largest, then silt, and clay grains are the smallest.

Diagram: the parts and what they do



Soil (مٹي · مٹی)
The soft layer made from broken rock, with mineral grains, humus, air, water and living things.

Sandy soil (واريءَ واري مٹي · ریتلی مٹی)
Large gritty grains. Water drains away fast. Found in deserts.

Silty soil (گاد واري مٹي · گادی مٹی)
Smooth silky grains. Holds water well and is fertile. Found along rivers.

Clay soil (چيڪي مٹي · چکنی مٹی)
The tiniest grains. Sticky when wet, hard when dry. Holds a lot of water.

Loam (لوم · میرا مٹی)
A mix of sand, silt, clay and humus. The best soil for plants.

4 Life in the soil and soil erosion

Soil is home to many living things. A single spoonful of soil can hold millions of microbes.

- **Bacteria and fungi** rot dead plants and animals and return **nutrients** to the soil. They are called **decomposers**.
- **Earthworms** dig tunnels that let air and water into the soil. They mix the soil and make it rich.
- Ants, insects and moles also live in the soil.

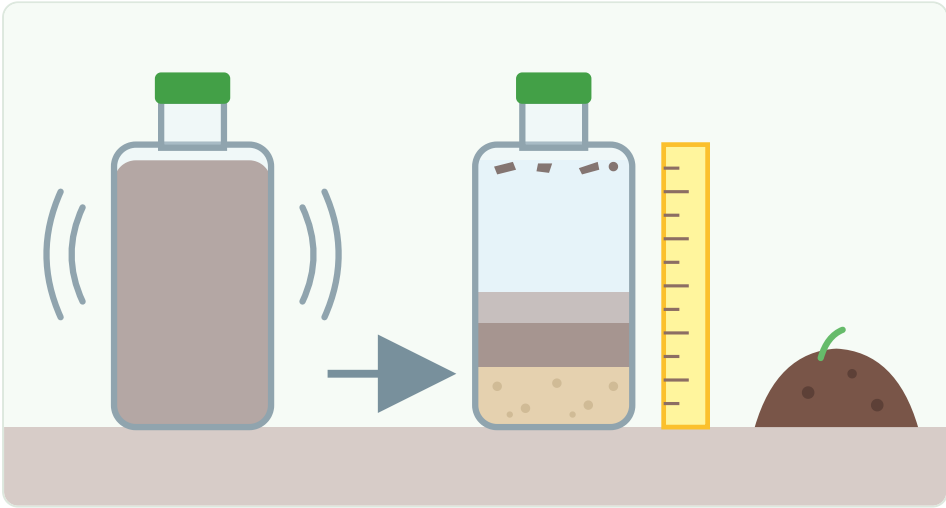
Running water and strong wind can carry away the rich top layer of soil. This is called **soil erosion**. Planting trees and grass helps, because roots hold the soil in place.

Key words

English	سنڌي	اردو
soil	مٹي	مٹی

English	سنڌي	اردو
sand	ريتي	ریت
silt	گاد	گاد
clay	چيڪي مٽي	چکنی مٹی
loam	لوم	میرا مٹی
humus	هيومس	ہیومس
minerals	معدنيات	معدنیات
decomposer	سڙائيندڙ	تحلیل کار
earthworm	ڪنڊيو	کیچوا
erosion	مٽيءَ جو کاتو	مٹی کا کٹاؤ

Experiment: What is in soil?



You need

- A clear plastic bottle or jar with a lid
- Two handfuls of garden soil
- Water
- A ruler

Steps

1. Fill one third of the bottle with soil.
2. Add water until the bottle is almost full, and close the lid tightly.
3. Shake the bottle well for one minute.
4. Leave the bottle still for one day.
5. Look at the layers and measure each layer with the ruler.

Safety: Use a plastic bottle instead of glass. Do not put soil in your mouth, and wash your hands with soap afterwards.

What did we learn? The soil settled in layers: the heaviest sand at the bottom, silt above it and fine clay on top. Bits of rotted leaves (humus) floated on the water. This shows that soil is a mixture of many things.

Practice questions

Circle the correct answer

1. **What is soil made from?**

   (a) Broken-down rocks **(b)** Plastic
   (c) Only water **(d)** Clouds

2. **What is the part of soil made from rotted leaves and dead animals called?**

   (a) Sand **(b)** Humus
   (c) Clay **(d)** Rock

3. **How does wet clay feel when you rub it between your fingers?**

   (a) Gritty **(b)** Sticky
   (c) Hot **(d)** Dry

4. **What does dark black soil usually have a lot of?**

   (a) Humus **(b)** Plastic
   (c) Salt **(d)** Glass

5. **Which soil is the best for plants?**

   (a) Sandy soil **(b)** Clay soil
   (c) Loam **(d)** Rock

6. **Which soil lets water drain away fastest?**

   (a) Clay soil **(b)** Sandy soil
   (c) Loam **(d)** Silty soil

7. **Which soil has the smallest grains?**

   (a) Sand **(b)** Silt
   (c) Clay **(d)** All are equal

8. **How do earthworms help the soil?**

   (a) They dig tunnels that let air and water in **(b)** They make the soil hard
   (c) They eat all the plants **(d)** They dry up the water

9. **What is it called when running water and wind carry away topsoil?**

   (a) Germination **(b)** Pollution
   (c) Soil erosion **(d)** Recycling

Write true or false

10. **Soil also contains air and water.** _____

11. **Roots grow easily in hard, tightly packed soil.** _____

12. **Clay soil becomes hard when it dries.** _____

13. **No living things live in soil.** _____

14. **Planting trees reduces soil erosion.** _____

Match the columns

15. **Match each soil with what it is like.**

1. Sandy soil		(a) Water stands on it
2. Clay soil		(b) Best for plants
3. Loam		(c) Drains fast

Match the columns

16. **Match each living thing with its job in the soil.**

1. Earthworm		(a) Rot dead things
2. Bacteria		(b) Digs tunnels

Worksheet: Soils

Name: _____ Date: _____

A. Fill in the blanks

Word box: Earthworms · Loam · erosion · humus · rocks

1. Soil is made from broken-down _____.
2. The rich part of soil made from rotted leaves is called _____.
3. _____ is the best soil for plants.
4. _____ dig tunnels that let air into the soil.
5. Soil being carried away by water and wind is called _____.

B. Put ✓ for true and ✗ for false

1. ☐ Sand grains are bigger than clay grains.
2. ☐ Sandy soil holds a lot of water.
3. ☐ Decomposers break down dead things into the soil.
4. ☐ Cutting down trees reduces soil erosion.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Sand	( ) Sticky
2. Silt	( ) Rotted leaves
3. Clay	( ) Gritty
4. Humus	( ) Smooth and silky

D. Answer in short

1. Name four things found in soil.

2. Why is loam the best soil for plants?

3. How can we stop soil erosion?

E. Draw and label

Draw the soil-in-a-bottle test and label the layers of sand, silt, clay and humus.

CHAPTER 10

Earth and Space

What we will learn Learn about the solar system, how day and night happen, what a year is, and why the moon seems to change shape.

1 The solar system

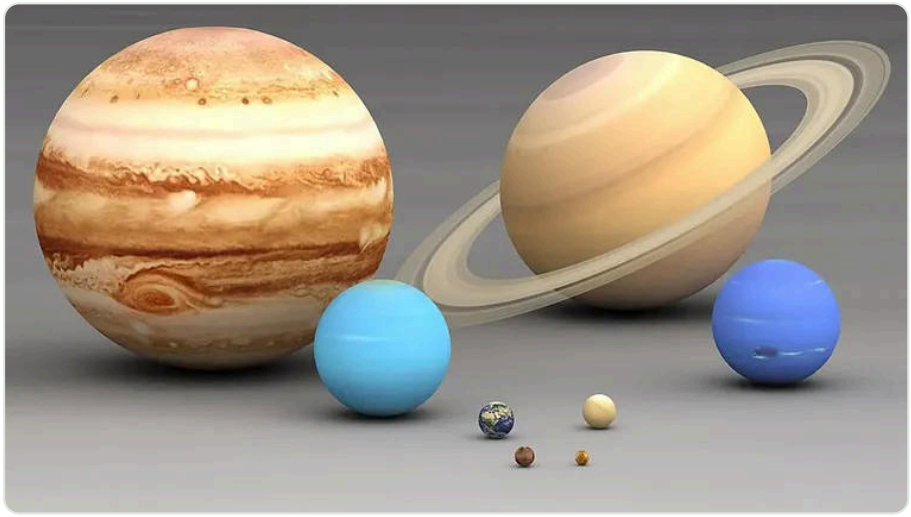
The **sun** is a **star**. It is a huge ball of hot gases that gives out light and heat.

The sun and the **planets**, moons and other objects that go around it make up the **solar system**.

Starting from the sun, the eight planets are: **Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune**.

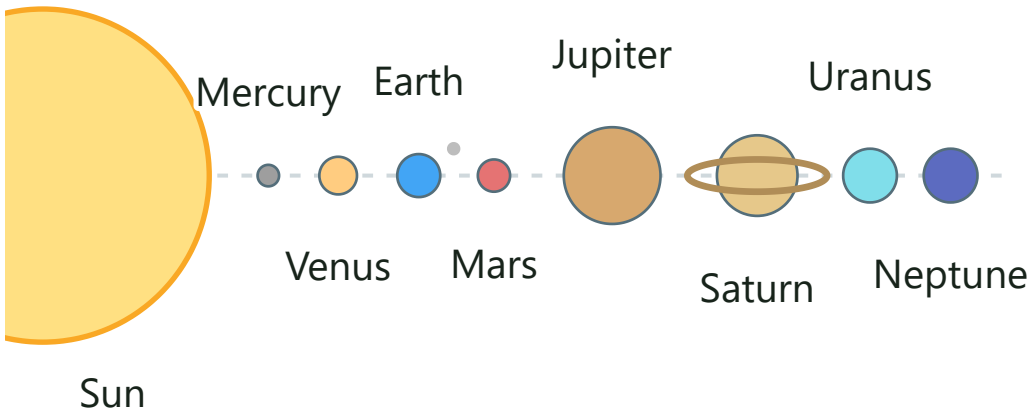
Our **Earth** is the only planet where we know there is life, because it has water, air and the right amount of heat.

Jupiter is the biggest planet and Mercury is the closest to the sun.



Picture: Lsmpascal · CC BY-SA 3.0 · Wikimedia Commons

Diagram: the parts and what they do



Sun (سج · سورج)
A star that gives light and heat to the solar system.

Mercury (عطارد · عطارد)
The closest planet to the sun and the smallest.

Venus (زهره · زہرہ)
The hottest planet. It shines like a bright star in the morning or evening.

Earth (زمين · زمین)
Our home: the planet with water, air and life. It has one moon.

Mars (مريخ · مریخ)
The red planet, because its soil has rust in it.

Jupiter (مشتري · مشتری)
The biggest planet, made mostly of gases.

Saturn (زحل · زحل)
It has beautiful rings made of ice and rock.

Uranus (يورينس · یورینس)
A very cold planet that spins lying on its side.

Neptune (نيپچون · نیپچون)
The farthest planet from the sun, with very strong winds.

2 Day, night and year

The Earth spins like a top on its **axis** (an imaginary straight line). This is called **rotation**. One full spin takes about **24 hours**.

The side of the Earth facing the sun has **day**, and the side facing away from the sun has **night**.

The Earth also travels around the sun. One full trip around the sun takes about **365 days** — one **year**.

The sun seems to rise in the **east** and set in the **west** every day, because the Earth is spinning.

3 The moon

The **moon** is the Earth's natural **satellite**, which means it goes around the Earth. One trip takes about a **month**.

The moon does not give out its own light. It **reflects** sunlight.

Half of the moon is always lit, but from the Earth we see different amounts of the lit half. That is why the moon seems to change **shape**: **new moon, crescent, half moon and full moon**.

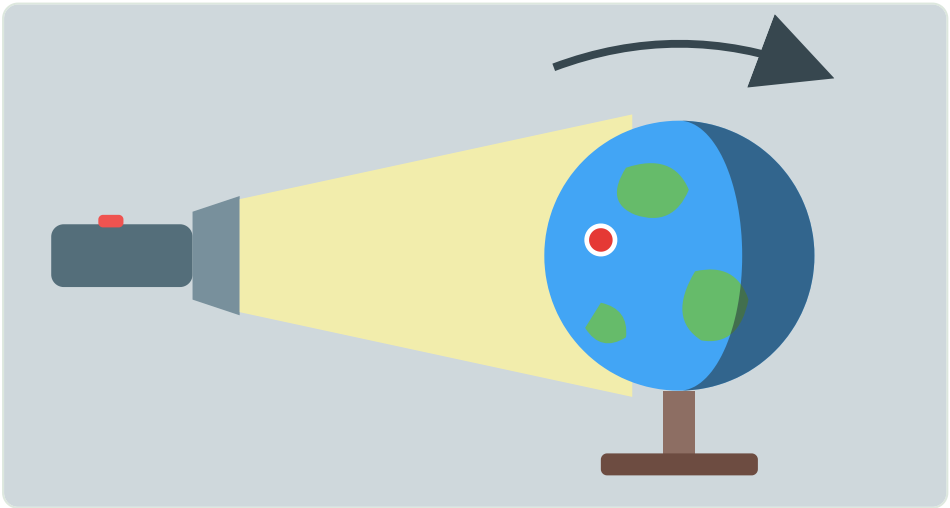
Islamic months begin with the moon. A thin crescent is looked for at the start of each new month.

There is no air or water on the moon, so nothing lives there.

Key words

English	سنڌي	اردو
solar system	شمسي نظام	نظامِ شمسی
star	تارو	ستارہ
planet	سيارو	سیارہ
satellite	سيٽلائيٽ	سیارچہ
axis	محور	محور
rotation	گردش	محوری گردش
revolution	سج جو چڪر	سالانہ گردش
east	اوپر	مشرق
west	اولهه	مغرب
crescent	هلال	ہلال
full moon	پورو چنڊ	پورا چاند

Experiment: Day and night with a torch and a ball



You need

- A torch (the sun)

Steps

1. Put a small mark on the ball. This is your home.

- A ball or an orange (the Earth)
- A small sticker or a pen mark
- A dark room

2. Turn off the room lights and ask a friend to shine the torch at the ball.
3. Slowly spin the ball.
4. Watch when the mark is in the light and when it is in the dark.

Safety: Move carefully in the dark and never shine the torch into anyone's eyes. Never look straight at the sun.

What did we learn? When the mark faces the torch it is in the light (day). As the ball spins and the mark moves to the other side, it is in the dark (night). Day and night happen because the Earth spins.

Practice questions

Circle the correct answer

1. **What is the sun?**

   (a) A star **(b)** A planet
   (c) A moon **(d)** A cloud

2. **How many planets are there in the solar system?**

   (a) 6 **(b)** 8
   (c) 10 **(d)** 12

3. **Which is the biggest planet?**

   (a) Earth **(b)** Mars
   (c) Jupiter **(d)** Mercury

4. **Which planet is closest to the sun?**

   (a) Mercury **(b)** Venus
   (c) Neptune **(d)** Saturn

5. **Why do we have day and night?**

   (a) The Earth spins on its axis **(b)** The sun switches off
   (c) The moon covers the sun **(d)** Clouds hide the sun

6. **How long does the Earth take to spin once on its axis?**

   (a) About 24 hours **(b)** One week
   (c) One month **(d)** 365 days

7. **How long does the Earth take to go once around the sun?**

(a) 24 hours **(b)** 30 days

(c) About 365 days **(d)** 10 years

8. **In which direction does the sun seem to rise?**

(a) East **(b)** West

(c) North **(d)** South

9. **What is the moon?**

(a) The Earth's natural satellite **(b)** A star

(c) A piece of the sun **(d)** A planet

Write true or false

10. **Earth is the only planet where we know there is life.** _____

11. **The moon makes its own light.** _____

12. **It is day on the side of the Earth that faces the sun.** _____

13. **There is lots of air and water on the moon.** _____

14. **Saturn has rings around it.** _____

Match the columns

15. Match each word with its meaning.

1. Rotation		(a) One trip around the sun
2. Year		(b) Something that goes around a planet
3. Satellite		(c) Earth spinning on its axis

Match the columns

16. Match each planet with a fact about it.

1. Mars		(a) The biggest planet
2. Jupiter		(b) Has beautiful rings
3. Saturn		(c) Has life
4. Earth		(d) The red planet

Worksheet: Earth and Space

Name: _____ Date: _____

A. Fill in the blanks

Word box: 24 · east · eight · satellite · star

1. The sun is a _____ .
2. There are _____ planets in the solar system.
3. The Earth spins once on its axis in about _____ hours.
4. The moon is the Earth's natural _____ .
5. The sun seems to rise in the _____ .

B. Put ✓ for true and ✗ for false

1. ☐ Jupiter is the smallest planet.
2. ☐ The Earth goes around the sun.
3. ☐ The moon reflects sunlight.
4. ☐ A year has about 100 days.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Mercury	( ) Planet with rings
2. Mars	( ) Round bright moon
3. Saturn	( ) Closest to the sun
4. Full moon	( ) Red planet

D. Answer in short

1. Write the names of the eight planets in order from the sun.

2. Why do day and night happen?

3. Why is life possible on Earth?

E. Draw and label

Draw four shapes of the moon: new moon, crescent, half moon and full moon, and label them.

CHAPTER 11

Plants: Parts and Making Food

What we will learn Learn the parts of a plant, their jobs, and how leaves make food.

1 Parts of a plant

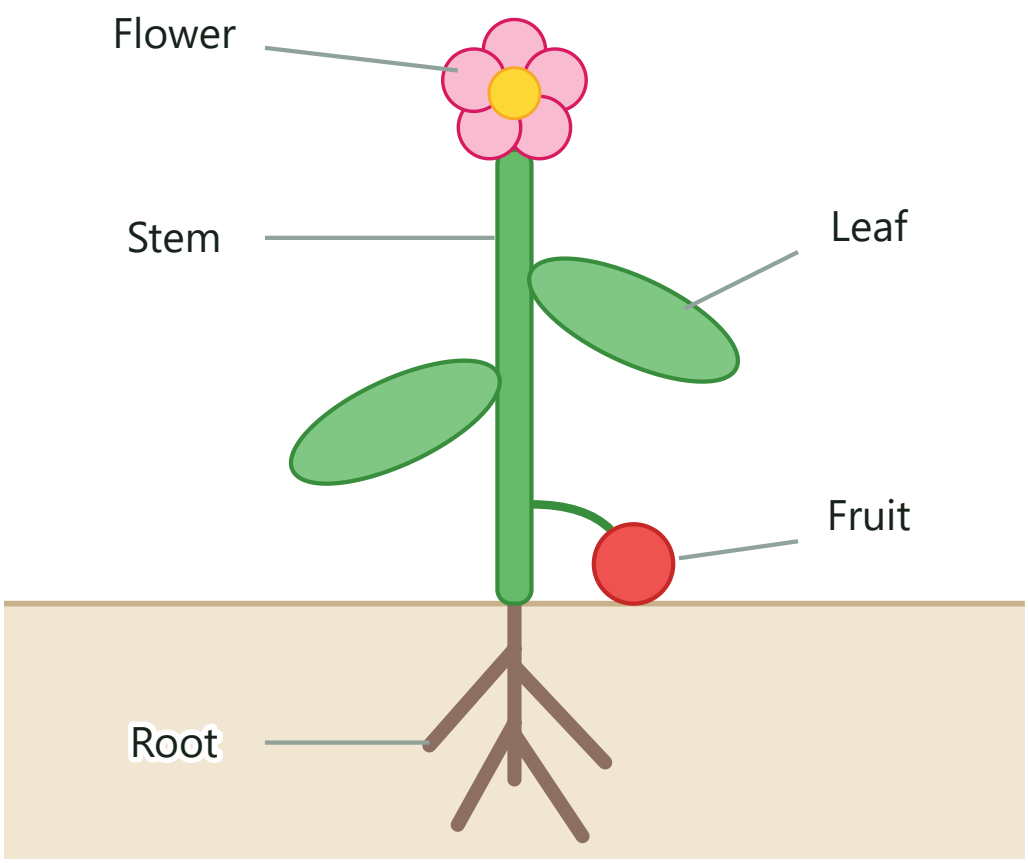
A plant has different **parts**. Each part has its own job.

- **Root**: grows under the soil. It holds the plant firmly and takes in **water** and minerals from the soil.
- **Stem**: holds the plant up. It carries water from the roots to the leaves, and food from the leaves to other parts.
- **Leaf**: is usually green and thin. Leaves make **food** for the plant.
- **Flower**: is often colourful. Flowers make **seeds**.
- **Fruit**: grows from the flower and keeps the seeds safe.
- **Seed**: can grow into a new plant.



Picture: Kelvinsong (labels removed) · CC BY-SA 3.0 · Wikimedia Commons

Diagram: the parts and what they do



Flower (گل · پھول)
Makes seeds. Its bright colours attract bees and butterflies.

Leaf (پن · پتا)
Makes food for the plant using sunlight, and gives out oxygen.

Stem (ڈنڈ · تنا)
Holds the plant up and carries water and food.

Fruit (میوو · پھل)
Grows from the flower and keeps the seeds safe.

Root (پاڑ · جڑ)
Holds the plant in the soil and takes in water and minerals.

2 How do plants make food?

Green plants make their own food. This process is called **photosynthesis**.

To make food, a leaf needs four things:

- **Sunlight** (energy from the sun)
- **Water** (from the roots)
- **Carbon dioxide** (a gas from the air that enters through tiny holes in the leaf)
- **Chlorophyll** (the green colour in leaves that traps sunlight)

The leaf makes **sugar (food)** and gives out **oxygen** into the air. We need this oxygen to breathe!

3 Why are plants important?

Plants are very important for all living things.

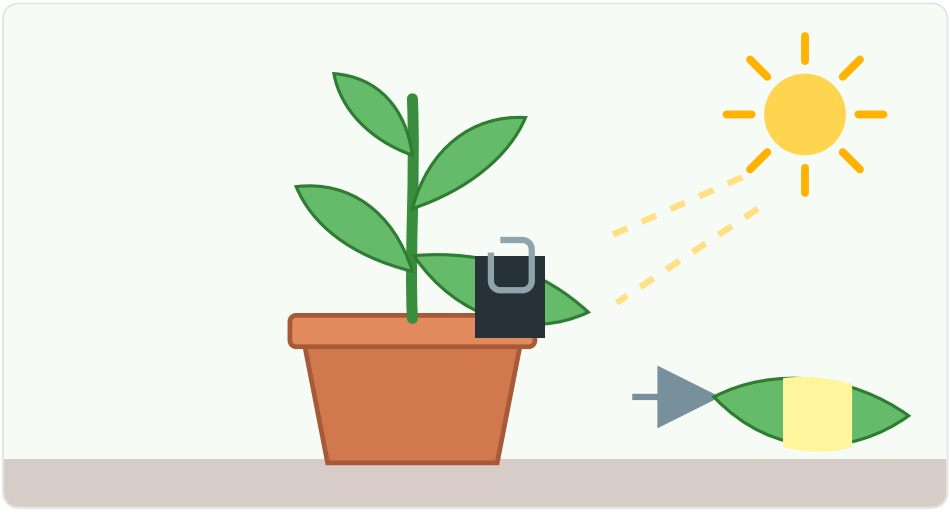
- They give us **food**: wheat, rice, fruits and vegetables.
- They give out **oxygen** for us to breathe.
- They give us **wood**, cotton for clothes, and medicines.
- Trees give **shade** and homes to birds and insects.

We should **plant trees** and take care of them.

Key words

English	سنڌي	اردو
plant	ٻوٽو	پودا
root	پاڙ	جڑ
stem	ٽڙ	تنا
leaf	پن	پتا
flower	گل	پھول
fruit	ميوو	پھل
seed	ٻج	بیج
photosynthesis	ضيائي تاليف	ضیائی تالیف
chlorophyll	ڪلوروفل	کلوروفل
oxygen	آڪسيجن	آکسیجن
carbon dioxide	ڪاربان ڊاءِ آڪسائيڊ	کاربن ڈائی آکسائیڈ
minerals	معدنيات	معدنیات

Experiment: Do leaves need sunlight?



You need

- A potted plant with green leaves
- Two small pieces of black or thick paper
- Two paper clips

Steps

1. Choose one leaf. Do not pluck it from the plant.
2. Cover part of the leaf with paper on both sides and fix it with the clips.
3. Keep the plant in sunlight for 4 to 5 days and keep watering it.
4. Remove the paper and compare the covered part with the rest of the leaf.

Safety: Ask an adult before you use a plant. Handle the leaf gently.

What did we learn? The covered part turns pale yellow, while the rest of the leaf stays green. Without sunlight the leaf cannot make chlorophyll and food. Leaves need sunlight to make food.

Practice questions

Circle the correct answer

1. **Which part of a plant takes in water from the soil?**

   (a) Leaf (b) Flower
   (c) Root (d) Fruit

2. **Which part holds the plant up and carries water?**

   (a) Stem (b) Seed
   (c) Flower (d) Leaf

3. **Which part of a plant makes seeds?**

   (a) Root (b) Flower
   (c) Stem (d) Leaf

4. **Which part keeps the seeds safe?**

   (a) Fruit (b) Root
   (c) Stem (d) Leaf

5. **Which part of a plant makes food?**

   (a) Root (b) Leaf
   (c) Seed (d) Stem

6. **What is the green colour in leaves called?**

   (a) Oxygen (b) Chlorophyll
   (c) Sugar (d) Soil

7. **Which gas do plants take in from the air to make food?**

(a) Oxygen (b) Carbon dioxide

(c) Smoke (d) Steam

8. **Which gas do plants give out that we breathe in?**

(a) Oxygen (b) Carbon dioxide

(c) Smoke (d) Dust

9. **Which of these do we get from plants?**

(a) Cotton (b) Glass

(c) Iron (d) Plastic

Write true or false

10. **Plants need sunlight to make food.** _____

11. **Roots usually grow above the soil.** _____

12. **A seed can grow into a new plant.** _____

13. **We get wood from trees.** _____

14. **Plants make food in the dark without any light.** _____

Match the columns

15. Match each plant part with its job.

1. Root		(a) Carries water up
2. Stem		(b) Makes food
3. Leaf		(c) Makes seeds
4. Flower		(d) Takes in water

Match the columns

16. Match each word with its meaning.

1. Sunlight		(a) Green colour in leaves
2. Chlorophyll		(b) Gas taken from the air
3. Carbon dioxide		(c) Gas given out by leaves
4. Oxygen		(d) Energy to make food

Worksheet: Plants: Parts and Making Food

Name: _____ Date: _____

A. Fill in the blanks

Word box: chlorophyll · food · oxygen · root · seeds

1. The _____ takes in water from the soil.
2. Leaves make _____ for the plant.
3. The green colour in leaves is called _____ .
4. Plants give out _____ into the air.
5. Flowers make _____ .

B. Put ✓ for true and ✗ for false

1. ☐ The stem carries water from the roots to the leaves.
2. ☐ Plants make food only at night.
3. ☐ We get cotton from plants.
4. ☐ A fruit grows from the root.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Leaf	( ) Grows into a new plant
2. Fruit	( ) Holds the plant up
3. Seed	( ) Makes food
4. Stem	( ) Keeps seeds safe

D. Answer in short

1. What four things does a leaf need to make food?

_____

_____

2. Write two jobs of the root.

_____

_____

3. Why should we plant trees?

_____

_____

E. Draw and label

Draw a plant and label the root, stem, leaf, flower and fruit.

CHAPTER 12

Our Body: The Digestive System

What we will learn Find out what happens to food after we eat it, and how to keep our teeth and stomach healthy.

1 Why do we need food?

Our body needs **food** to get **energy**, to grow and to stay healthy.

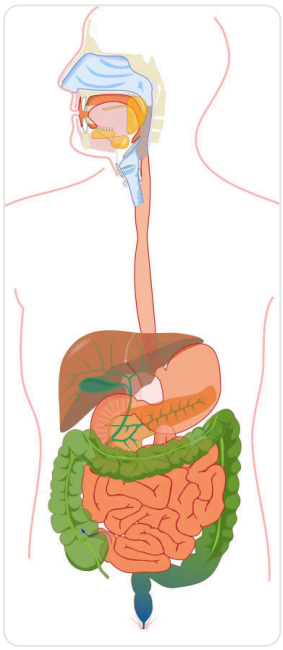
But the body cannot use food as it is. Food must be broken down into very small pieces. This is called **digestion**.

The parts of the body that do this work together are called the **digestive system**.

2 The journey of food

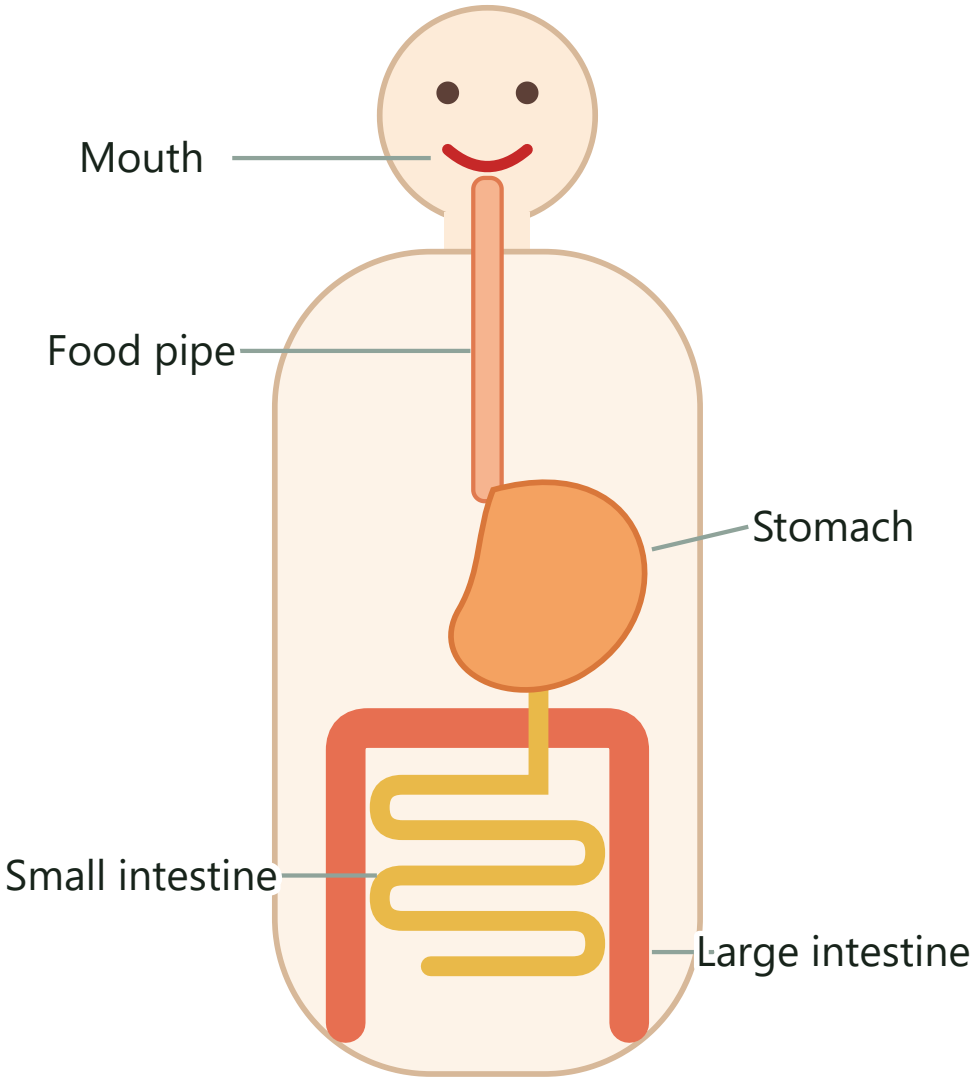
Let us follow a bite of roti on its journey through the body.

- **Mouth**: Teeth cut and chew the food. **Saliva** makes it wet and soft and starts digestion.
- **Food pipe**: A long tube that carries food from the mouth to the stomach.
- **Stomach**: A bag made of muscles. It mixes the food well and adds digestive juices to it.
- **Small intestine**: A long, coiled tube. Here digestion finishes and useful food passes into the **blood**.
- **Large intestine**: It takes water back from the leftover food. Then the waste leaves the body.



Picture: Mariana Ruiz, Jmarchn · Public domain · Wikimedia Commons

Diagram: the parts and what they do



Mouth (وات · منہ)
Teeth chew the food and saliva makes it soft. Digestion starts here.

Food pipe (کاڈي جي نالي · غذا کی نالی)
A long tube that carries food from the mouth to the stomach.

Stomach (معدو · معدہ)
A bag of muscles that mixes food well and adds digestive juices.

Small intestine (ننڍو آنڊو · چھوٹی آنت)
A long coiled tube where digestion finishes and food goes into the blood.

Large intestine (وڏو آنڊو · بڑی آنت)
Takes water back from leftover food. Then the waste leaves the body.

3 Our teeth

Teeth help us cut and chew food. We have different kinds of teeth:

- **Front teeth (incisors)** are thin and sharp. They **cut** food.
- **Pointed teeth (canines)** **tear** food.

- **Molars** are wide back teeth. They **grind** and chew food.

Children have 20 **milk teeth**. They fall out and **permanent teeth** grow in their place.

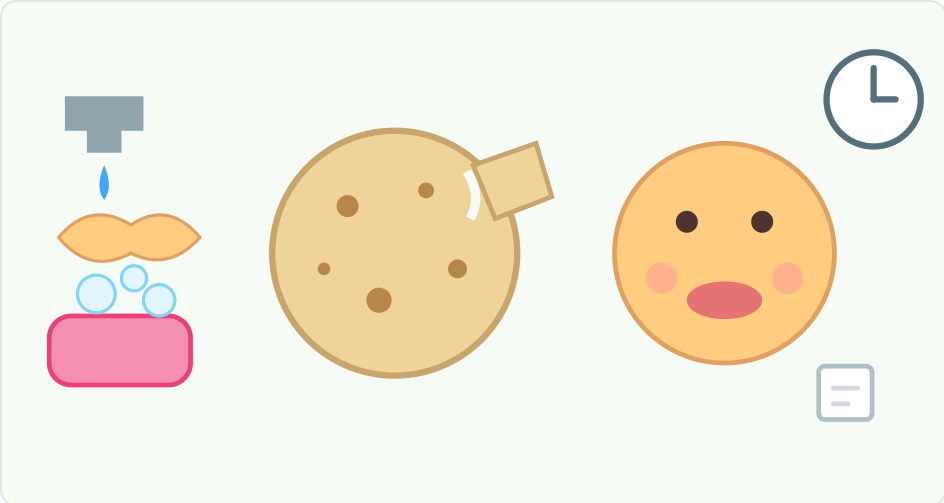
4 Healthy habits

- **Brush** your teeth twice a day, in the morning and before sleeping.
- **Wash your hands** with soap before eating.
- **Chew** your food well and eat slowly.
- Eat fruits, vegetables and **clean food**. Eat fewer sweets and chips.
- Drink plenty of **clean water**.

Key words

English	سنڌي	اردو
digestion	هاضمو	ہاضمہ
digestive system	هاضمي جو نظام	نظامِ انہضام
energy	توانائي	توانائی
mouth	وات	منہ
saliva	لعاب	لعاب
food pipe	کاڌي جي نالي	غذا کی نالی
stomach	معدو	معدہ
small intestine	ننڍو آنڊو	چھوٹی آنت
large intestine	وڏو آنڊو	بڑی آنت
teeth	ڏند	دانت
molars	ڏاڙهون	داڑھیں

Experiment: Why does roti taste sweet?



You need

- A small piece of plain roti or bread
- Clean hands

Steps

1. Wash your hands with soap.
2. Put a small piece of roti in your mouth.
3. Chew it slowly for about one minute. Do not swallow it quickly.
4. Notice how the taste changes.

Safety: Use only clean, fresh food. Never share food that has been in your mouth.

What did we learn? After chewing for a while, the roti starts to taste a little sweet. Saliva changes some of the starch in the roti into sugar. This shows that digestion starts in the mouth!

Practice questions

Circle the correct answer

1. **What is breaking food down into very small pieces inside the body called?**

   (a) Digestion (b) Breathing
   (c) Growing (d) Sleeping

2. **Where does digestion start?**

   (a) In the stomach (b) In the mouth
   (c) In the large intestine (d) In the food pipe

3. **Which tube carries food from the mouth to the stomach?**

   (a) Food pipe (b) Small intestine
   (c) Wind pipe (d) Large intestine

4. **In which part does useful food pass into the blood?**

(a) Mouth (b) Small intestine

(c) Teeth (d) Food pipe

5. **Which part takes back water from leftover food?**

(a) Large intestine (b) Mouth

(c) Stomach (d) Teeth

6. **Which teeth grind food?**

(a) Front teeth (b) Pointed teeth

(c) Molars (d) None of them

7. **How many milk teeth do children have?**

(a) 10 (b) 20

(c) 32 (d) 40

8. **How often should we brush our teeth?**

(a) Once a week (b) Twice a day

(c) Never (d) Once a month

9. **What makes food wet and soft in the mouth?**

(a) Saliva (b) Blood

(c) Air (d) Bones

Write true or false

10. **The stomach is a bag made of muscles.** _____

11. **Pointed teeth (canines) tear food.** _____

12. **We should wash our hands before eating.** _____

13. **Eating lots of sweets is good for our teeth.** _____

14. **Food goes from the mouth straight to the large intestine.** _____

Match the columns

15. Match each body part with its job.

1. Mouth		(a) Carries food to the stomach
2. Food pipe		(b) Mixes food with juices
3. Stomach		(c) Sends food into the blood
4. Small intestine		(d) Chews food

Match the columns

16. Match each type of tooth with its job.

1. Front teeth		(a) Tear
2. Pointed teeth		(b) Grind
3. Molars		(c) Fall out in childhood
4. Milk teeth		(d) Cut

Worksheet: Our Body: The Digestive System

Name: _____ Date: _____

A. Fill in the blanks

Word box: Molars · food pipe · mouth · small intestine · twice

1. Digestion starts in the _____ .
2. The _____ carries food from the mouth to the stomach.
3. Useful food goes into the blood in the _____ .
4. _____ grind and chew food.
5. We should brush our teeth _____ a day.

B. Put ✓ for true and ✗ for false

1. ☐ Saliva makes food soft.
2. ☐ Children have 20 milk teeth.
3. ☐ We should eat food without washing our hands.
4. ☐ The stomach mixes food with digestive juices.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Front teeth	( ) Grind food
2. Pointed teeth	( ) Takes back water
3. Molars	( ) Cut food
4. Large intestine	( ) Tear food

D. Answer in short

1. Why does our body need food?

_____

_____

2. Write the parts of the digestive system in order.

_____

_____

3. Write two healthy habits for good digestion.

_____

_____

E. Draw and label

Draw the digestive system and label the mouth, food pipe, stomach, small intestine and large intestine.

CHAPTER 13

Water and the Water Cycle

What we will learn Learn the three states of water, the water cycle and ways to save water.

1 The three states of water

Water can be found in three forms, called **states**:

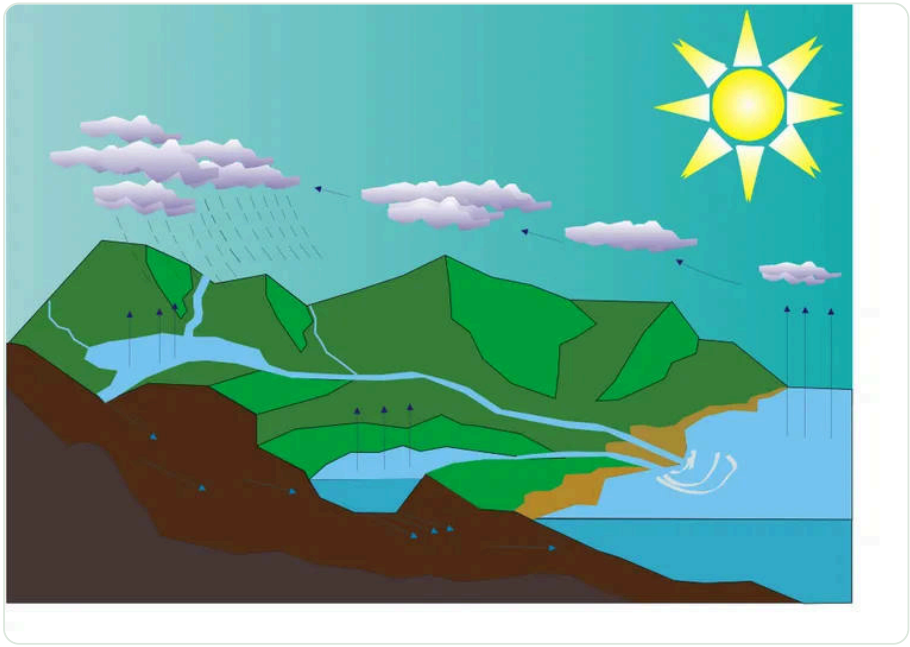
- **Solid**: ice. It has a fixed shape.
- **Liquid**: water. It flows and takes the shape of its container.
- **Gas**: water vapour. We cannot see it. It spreads out in the air.

When ice is heated, it **melts** into water. When water is made very cold, it **freezes** into ice.

2 The water cycle

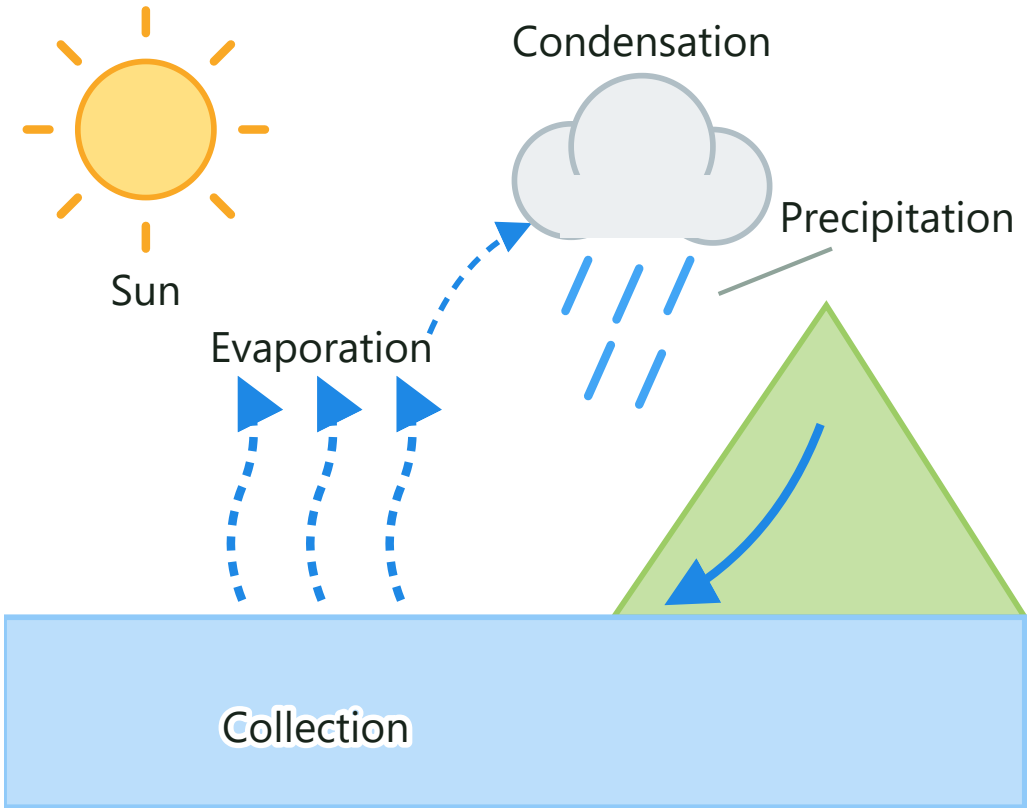
Water on Earth moves round and round. This is called the **water cycle**. It has four steps:

- **Evaporation**: The sun heats water in seas, rivers and ponds. The water turns into water vapour and rises up.
- **Condensation**: High up, the air is cool. Water vapour cools down and turns into tiny drops. These drops make **clouds**.
- **Precipitation**: When the drops become big and heavy, they fall as **rain** (or snow or hail).
- **Collection**: Rain water collects in rivers, lakes, seas and under the ground. Then the cycle starts again.



Picture: Ingwik · CC BY-SA 3.0 · Wikimedia Commons

Diagram: the parts and what they do



Sun (سج · سورج)
The sun heats the water and starts the water cycle.

Evaporation (ٻاڦ ٿيڻ · تبخير)
Heat from the sun turns water into water vapour, which rises up.

Condensation (ڪکر ٿهڻ · تکثيف)
High up, water vapour cools into tiny drops that form clouds.

Precipitation (مينهن وسڻ · بارش)
When drops get big and heavy, they fall as rain, snow or hail.

Collection (پاڻي گڏ ٿيڻ · جمع ہونا)
Water collects in rivers, lakes, seas and under the ground, then the cycle starts again.

3 Let us save water

Only a small part of the water on Earth is **clean water** that we can drink. We must not waste it.

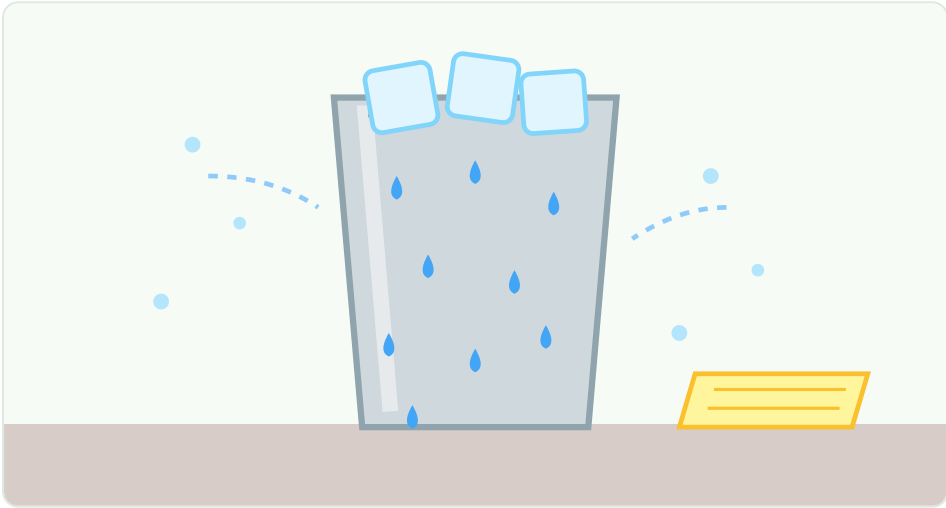
- **Turn off the tap** while brushing your teeth.
- Fix **leaking** taps and pipes.
- Water plants in the **morning or evening**, when less water evaporates.
- Do not throw rubbish into rivers, canals and ponds. Keep water **clean**.

Key words

English	سنڌي	اردو
water	پاڻي	پانی
solid	ٺوس	ٹھوس

English	سنڌي	اردو
liquid	مائع	مائع
gas	گئس	گیس
water vapour	ٻاڦ	آبی بخارات
melting	ڳرڻ	پگھلنا
freezing	جمڻ	جمنا
water cycle	پاڻيءَ جو چڪر	آبی چکر
evaporation	ٻاڦ ٿيڻ	تبخیر
condensation	ڪڪر ٺهڻ	تکثیف
precipitation	مينهن وسڻ	بارش (ترسیب)
cloud	ڪڪر	بادل

Experiment: Water drops from the air



You need

- A steel glass or bottle
- Some ice cubes (or very cold water from the fridge)
- A dry cloth

Steps

1. Wipe the outside of the glass with the dry cloth.
2. Fill the glass with ice and a little water.
3. Wait for 5 minutes and do not touch the glass.
4. Look at the outside of the glass. What do you see?

Safety: Be careful not to drop the glass. Wipe up any water on the floor so no one slips.

What did we learn? Small drops of water appear on the outside of the glass. They come from water vapour in the air, which cools on the cold glass and turns into drops. This is condensation — clouds form in the sky in the same way!

Practice questions

Circle the correct answer

1. **Ice is water in which state?**

   (a) Solid **(b)** Liquid
   (c) Gas **(d)** None of these

2. **What happens when ice is heated?**

   (a) It freezes **(b)** It melts
   (c) It burns **(d)** It grows

3. **Water vapour is water in which state?**

   (a) Solid **(b)** Liquid
   (c) Gas **(d)** Ice

4. **The sun heats water and it turns into vapour. What is this called?**

   (a) Evaporation **(b)** Condensation
   (c) Precipitation **(d)** Collection

5. **What are clouds made of?**

   (a) Tiny drops of water **(b)** Smoke
   (c) Cotton **(d)** Sand

6. **Water vapour cools and turns into tiny drops. What is this called?**

   (a) Evaporation **(b)** Condensation
   (c) Melting **(d)** Collection

7. **What is rain falling from clouds called?**

   (a) Precipitation **(b)** Evaporation
   (c) Freezing **(d)** Condensation

8. **What gives the heat that starts the water cycle?**

   (a) The moon **(b)** The sun
   (c) The wind **(d)** The stars

9. **Which is a good way to save water?**

   (a) Leave the tap running **(b)** Turn off the tap while brushing
   (c) Throw rubbish in canals **(d)** Wash the car every day with a pipe

Write true or false

10. **Liquid water takes the shape of its container.** _____

11. **We can easily see water vapour in the air.** _____

12. **Rain water collects in rivers, lakes and seas.** _____

13. **The water cycle happens only once and then stops.** _____

14. **We should throw rubbish into rivers.** _____

Match the columns

15. **Match each step of the water cycle with what happens.**

1. Evaporation		(a) Clouds form
2. Condensation		(b) Rain falls
3. Precipitation		(c) Water gathers in seas and rivers
4. Collection		(d) Water turns into vapour

Match the columns

16. **Match each word with its state or change.**

1. Ice		(a) Liquid
2. Water		(b) Gas
3. Water vapour		(c) Ice turning into water
4. Melting		(d) Solid

Worksheet: Water and the Water Cycle

Name: _____ Date: _____

A. Fill in the blanks

Word box: evaporation · solid · sun · tap · water

1. Ice is water in the _____ state.
2. When water turns into vapour, it is called _____ .
3. Clouds are made of tiny drops of _____ .
4. The _____ heats water in seas and rivers.
5. We should turn off the _____ while brushing.

B. Put ✓ for true and ✗ for false

1. ☐ Water vapour is a gas.
2. ☐ Ice has a fixed shape.
3. ☐ Clouds are made of smoke.
4. ☐ We should fix leaking taps.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Solid	( ) Air
2. Liquid	( ) Water turning into ice
3. Gas	( ) Ice
4. Freezing	( ) Milk

D. Answer in short

1. Name the four steps of the water cycle.

_____

_____

2. What happens to ice when it is heated?

_____

_____

3. Write two ways to save water at home.

_____

_____

E. Draw and label

Draw the water cycle. Show the sun, clouds, rain and a river or sea, and label each step.

FOR TEACHERS

Answer key

Chapter 1: Living Things and Classification

Practice questions

1. (c) Tree
2. (a) Food and water
3. (b) Reproduction
4. (a) Pencil
5. (b) Sand
6. (a) Classification
7. (b) Plants
8. (c) Fish
9. (b) Birds
10. (a) Amphibians
11. False
12. True
13. True
14. True
15. 1 → (d), 2 → (a), 3 → (b), 4 → (c)
16. 1 → (d), 2 → (a), 3 → (b), 4 → (c)

Worksheet

A. Fill in the blanks: 1. water · 2. classification · 3. vertebrates · 4. feathers · 5. food

B. Put ✓ for true and ✗ for false: 1. ✗ · 2. ✓ · 3. ✗ · 4. ✓

C. Match column A with column B (write the number in the brackets): (3) Non-flowering plant · (4) Reptile · (1) Invertebrate · (2) Flowering plant

D. Answer in short:

1. They grow, they breathe, and they need food and water.
2. It cannot grow, breathe, eat or reproduce. It moves only when someone drives it.
3. Fish, amphibians, reptiles, birds and mammals.

Chapter 2: Microorganisms

Practice questions

1. (a) A microscope
2. (b) Everywhere: in soil, air, water and our bodies
3. (c) Viruses
4. (a) A fungus
5. (b) Yeast
6. (a) A fungus
7. (b) Decomposition
8. (b) Mosquito
9. (a) Washing hands with soap before eating
10. False
11. True
12. True
13. False
14. True
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (c), 2 → (a), 3 → (b)

Worksheet

A. Fill in the blanks: 1. microscope · 2. Viruses · 3. Yeast · 4. fungus · 5. infection

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Common cold · (4) Polio drops · (1) Naan and bread · (2) Yoghurt

D. Answer in short:

1. Bacteria, viruses and fungi.
2. Yeast makes dough rise; bacteria turn milk into yoghurt.
3. Wash hands with soap, boil drinking water and keep food covered.

Chapter 3: Seeds: Structure and Germination

Practice questions

1. (a) Seed coat
2. (b) Radicle
3. (c) Bean
4. (a) In the endosperm
5. (b) Two
6. (c) The radicle
7. (a) Bean
8. (d) Soil
9. (a) To breathe and get energy
10. True
11. False
12. True
13. False
14. True
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (b), 2 → (a)

Worksheet

A. Fill in the blanks: 1. seed coat · 2. plumule · 3. endosperm · 4. germination · 5. temperature

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✗ · 4. ✓

C. Match column A with column B (write the number in the brackets): (3) Monocot seed · (4) Dicot seed · (1) Root · (2) Stem and leaves

D. Answer in short:

1. Seed coat, embryo and cotyledons.
2. A bean has two cotyledons; maize has one cotyledon and an endosperm.
3. Water, air (oxygen) and the right temperature.

Chapter 4: Environmental Pollution

Practice questions

1. (a) Harmful things getting into air, water or land
2. (b) Pollutants
3. (c) Burning rubbish
4. (a) Asthma
5. (b) Sewage
6. (a) Cholera
7. (c) Banana peel
8. (b) A cloth bag
9. (a) Making new things from old things
10. True
11. False
12. True
13. False
14. True
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (b), 2 → (a)

Worksheet

A. Fill in the blanks: 1. pollutants · 2. sewage · 3. asthma · 4. Plastic · 5. recycling

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Plastic bags · (4) Vegetable peels · (1) Smoke from vehicles · (2) Factory chemicals

D. Answer in short:

1. Air pollution, water pollution and land pollution.
2. Biodegradable waste rots into the soil; non-biodegradable waste does not rot for many years.
3. Use cloth bags, do not burn rubbish and plant trees.

Chapter 5: Matter and Its Changes

Practice questions

1. (a) Anything that takes up space and has mass
2. (a) Air takes up space
3. (a) Solid
4. (b) Liquid
5. (c) Oxygen
6. (c) Paper burning
7. (c) Water freezing into ice
8. (a) A new substance is made
9. (a) Particles
10. False
11. True
12. False
13. True
14. False
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (b), 2 → (a)

Worksheet

A. Fill in the blanks: 1. matter · 2. solid · 3. container · 4. new · 5. rust

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Gas · (4) Chemical change · (1) Solid · (2) Liquid

D. Answer in short:

1. Solid (stone), liquid (water), gas (air).
2. No new substance is made in a physical change, but a new substance is made in a chemical change.
3. When we blow up a balloon, air fills it and it gets bigger.

Chapter 6: Force and Simple Machines

Practice questions

1. (a) A push or a pull
2. (b) The shape
3. (b) Because of gravity
4. (c) A rough stony road
5. (a) Lever
6. (b) Pulley
7. (c) Wedge
8. (a) Inclined plane
9. (b) Muscular force
10. True
11. False
12. True
13. False
14. True
15. 1 → (d), 2 → (a), 3 → (b), 4 → (c)
16. 1 → (c), 2 → (a), 3 → (b)

Worksheet

A. Fill in the blanks: 1. force · 2. gravity · 3. friction · 4. pulley · 5. lever

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Axe · (4) Ramp · (1) Cart · (2) Bottle cap

D. Answer in short:

1. A force can start an object moving and can change its shape.
2. Friction lets us walk without slipping.
3. Lever, pulley, wheel and axle, inclined plane, wedge and screw.

Chapter 7: Light and Sound

Practice questions

1. (b) The sun
2. (a) It reflects sunlight
3. (a) When light from it enters our eyes
4. (a) In straight lines
5. (a) When an opaque object blocks the light
6. (a) Transparent
7. (a) By vibrations
8. (c) In empty space
9. (c) A torch
10. False
11. True
12. True
13. False
14. True
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (c), 2 → (a), 3 → (b)

Worksheet

A. Fill in the blanks: 1. luminous · 2. straight · 3. shadow · 4. vibrations · 5. space

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Sound from vibration · (4) Translucent · (1) Luminous object · (2) Non-luminous object

D. Answer in short:

1. The sun, a bulb and a candle.
2. When an opaque object blocks light, a dark area forms behind it.
3. Solids, liquids and gases.

Chapter 8: Electricity and Magnets

Practice questions

1. (a) A battery
2. (c) Flying a kite in an open field away from wires
3. (a) When the circuit is complete (closed)
4. (c) Copper
5. (a) Plastic is an insulator and keeps us safe
6. (a) An iron nail
7. (a) They push each other away
8. (a) A compass
9. (a) Opens or closes the circuit
10. True
11. False
12. True
13. False
14. True
15. 1 → (d), 2 → (a), 3 → (b), 4 → (c)
16. 1 → (b), 2 → (a)

Worksheet

A. Fill in the blanks: 1. battery · 2. conductors · 3. switch · 4. poles · 5. compass

B. Put ✓ for true and ✗ for false: 1. ✗ · 2. ✓ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Gives light · (4) Push away · (1) Conductor · (2) Insulator

D. Answer in short:

1. Do not touch switches with wet hands and never push anything into a socket.
2. Battery, wires, switch and bulb.
3. In a compass, and to keep a fridge door shut.

Chapter 9: Soils

Practice questions

1. (a) Broken-down rocks
2. (b) Humus
3. (b) Sticky
4. (a) Humus
5. (c) Loam
6. (b) Sandy soil
7. (c) Clay
8. (a) They dig tunnels that let air and water in
9. (c) Soil erosion
10. True
11. False
12. True
13. False
14. True
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (b), 2 → (a)

Worksheet

A. Fill in the blanks: 1. rocks · 2. humus · 3. Loam · 4. Earthworms · 5. erosion

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Sticky · (4) Rotted leaves · (1) Gritty · (2) Smooth and silky

D. Answer in short:

1. Mineral grains, humus, air and water (and living things).
2. It has sand, silt, clay and humus, and holds the right amount of air and water.
3. By planting trees and grass, because roots hold the soil in place.

Chapter 10: Earth and Space

Practice questions

1. (a) A star
2. (b) 8
3. (c) Jupiter
4. (a) Mercury
5. (a) The Earth spins on its axis
6. (a) About 24 hours
7. (c) About 365 days
8. (a) East
9. (a) The Earth's natural satellite
10. True
11. False
12. True
13. False
14. True
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (d), 2 → (a), 3 → (b), 4 → (c)

Worksheet

A. Fill in the blanks: 1. star · 2. eight · 3. 24 · 4. satellite · 5. east

B. Put ✓ for true and ✗ for false: 1. ✗ · 2. ✓ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Planet with rings · (4) Round bright moon · (1) Closest to the sun · (2) Red planet

D. Answer in short:

1. Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune.
2. The Earth spins on its axis. The side facing the sun has day and the other side has night.
3. Because it has water, air and the right amount of heat.

Chapter 11: Plants: Parts and Making Food

Practice questions

1. (c) Root
2. (a) Stem
3. (b) Flower
4. (a) Fruit
5. (b) Leaf
6. (b) Chlorophyll
7. (b) Carbon dioxide
8. (a) Oxygen
9. (a) Cotton
10. True
11. False
12. True
13. True
14. False
15. 1 → (d), 2 → (a), 3 → (b), 4 → (c)
16. 1 → (d), 2 → (a), 3 → (b), 4 → (c)

Worksheet

A. Fill in the blanks: 1. root · 2. food · 3. chlorophyll · 4. oxygen · 5. seeds

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Grows into a new plant · (4) Holds the plant up · (1) Makes food · (2) Keeps seeds safe

D. Answer in short:

1. Sunlight, water, carbon dioxide and chlorophyll.
2. It holds the plant in the soil and takes in water.
3. Trees give us oxygen, food, wood and shade.

Chapter 12: Our Body: The Digestive System

Practice questions

1. (a) Digestion
2. (b) In the mouth
3. (a) Food pipe
4. (b) Small intestine
5. (a) Large intestine
6. (c) Molars
7. (b) 20
8. (b) Twice a day
9. (a) Saliva
10. True
11. True
12. True
13. False
14. False
15. 1 → (d), 2 → (a), 3 → (b), 4 → (c)
16. 1 → (d), 2 → (a), 3 → (b), 4 → (c)

Worksheet

A. Fill in the blanks: 1. mouth · 2. food pipe · 3. small intestine · 4. Molars · 5. twice

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✓ · 3. ✗ · 4. ✓

C. Match column A with column B (write the number in the brackets): (3) Grind food · (4) Takes back water · (1) Cut food · (2) Tear food

D. Answer in short:

1. To get energy, to grow and to stay healthy.
2. Mouth, food pipe, stomach, small intestine, large intestine.
3. Chew food well and wash your hands before eating.

Chapter 13: Water and the Water Cycle

Practice questions

1. (a) Solid
2. (b) It melts
3. (c) Gas
4. (a) Evaporation
5. (a) Tiny drops of water
6. (b) Condensation
7. (a) Precipitation
8. (b) The sun
9. (b) Turn off the tap while brushing
10. True
11. False
12. True
13. False
14. False
15. 1 → (d), 2 → (a), 3 → (b), 4 → (c)
16. 1 → (d), 2 → (a), 3 → (b), 4 → (c)

Worksheet

A. Fill in the blanks: 1. solid · 2. evaporation · 3. water · 4. sun · 5. tap

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✓ · 3. ✗ · 4. ✓

C. Match column A with column B (write the number in the brackets): (3) Air · (4) Water turning into ice · (1) Ice · (2) Milk

D. Answer in short:

1. Evaporation, condensation, precipitation and collection.
2. It melts and turns into water.
3. Turn off the tap while brushing and fix leaking taps.

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